

PEAO: A Cooperative Parallel Enzyme Optimization Algorithm with Adaptive Search Mechanisms

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Abstract

Bio-inspired optimization algorithms have become an effective class of techniques for addressing challenging continuous optimization problems. In this work, we introduce the Parallel Enzyme Action Optimization (PEAO) algorithm, a parallel bio-inspired optimization approach that incorporates a multi-strategy communication mechanism among cooperative subpopulations. The population is partitioned into multiple subpopulations that evolve concurrently, promoting a more effective exploration of the search space. Furthermore, PEAO integrates adaptive search factors, local search procedure and communication strategies to improve solution quality while reducing the risk of premature convergence. In addition, a K-means-based population initialization procedure and a convergence-driven stopping criterion based on successive improvements in the best objective-function value are incorporated to reduce unnecessary objective-function evaluations and improve the overall efficiency of the optimization process.

Keywords: Parallel optimization; Evolutionary techniques; Stochastic methods; high-dimensional problems; Multi-strategy communication

1. Introduction

Global optimization aims to identify the global optimum of a multidimensional objective function over a bounded search space. The optimization problem is formally defined as:

$$x^* = \arg \min_{x \in S} f(x) \quad (1)$$

with S :

$$S = [a_1, b_1] \times [a_2, b_2] \times \dots [a_n, b_n]$$

with a_i and b_i representing lower and upper bounds for each variable x_i .

The continuous increase in the dimensionality and complexity of modern optimization problems has created a growing demand for optimization algorithms capable of efficiently exploring large and multimodal search spaces [1–4] while maintaining reasonable computational requirements. Such challenges often arise in mechanical design [5,6] machine learning [7,8] and data analysis [9] where the search landscape often contains many local optima. Although population-based metaheuristic algorithms have demonstrated significant success in addressing these problems, their performance often deteriorates as the dimensionality

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of the problem increases, leading to slow convergence, premature stagnation and excessive computational cost. Recent surveys have highlighted the continuous development of bio-inspired and evolutionary optimization algorithms, emphasizing the importance of balancing exploration, exploitation, scalability and computational efficiency in modern optimization problems [10,11]. Moreover, recent adaptive evolutionary algorithms have demonstrated remarkable performance on large-scale optimization benchmarks, motivating the development of more efficient cooperative optimization frameworks [12,13]. The Enzyme Action Optimizer (EAO) is a recently introduced bio-inspired optimization algorithm that models the catalytic behavior of enzymes during biochemical reactions. Since its introduction by Rodan et al. [14] EAO has demonstrated competitive performance across a wide range of benchmark and real-world optimization problems. Recent studies have successfully applied EAO to infinite impulse response filter identification problems, where it exhibited strong convergence characteristics under different filter configurations [15]. Furthermore, modified variants of EAO have been employed for the design of fractional-order PID controllers in nonlinear continuous stirred tank reactor systems, demonstrating effective control performance and robustness in challenging dynamic environments [16]. Enhanced EAO frameworks have also been proposed for feature selection in medical data classification tasks, achieving improved predictive performance through more effective exploration and diversity preservation mechanisms [17]. In addition to these applications, EAO has been successfully utilized for the design of robust two-degree-of-freedom PIDA controllers in magnetic levitation systems, highlighting its effectiveness in complex control engineering problems [18]. The algorithm has also been applied to parameter identification in nonlinear systems, further demonstrating its capability to handle challenging estimation and modeling tasks [19]. Moreover, recent research has introduced the Dynamic Enzyme Action Optimizer (DEAO), an adaptive variant of EAO that incorporates dynamic search mechanisms to improve the balance between exploration and exploitation and enhance optimization performance on classical benchmark functions [20]. Collectively, these studies demonstrate the flexibility, robustness and adaptability of the EAO search mechanism across a diverse range of optimization applications. Despite these promising developments, the original EAO and most of its existing variants primarily rely on single-population search processes. Consequently, they may encounter difficulties when addressing large-scale and highly multimodal optimization problems where maintaining population diversity, preserving exploration capability and avoiding premature convergence become increasingly challenging. In addition, the absence of structured cooperation among search agents may limit the efficient dissemination of useful search information throughout the population. Conventional island-model evolutionary algorithms divide the population into multiple semi-independent subpopulations (islands) that evolve independently while periodically exchanging individuals according to predefined migration policies. These migration strategies are typically characterized by three main design aspects: the migration granularity (the number of exchanged individuals), the migration timing (the migration interval or frequency) and the migration content (elite, random or selected individuals). The effectiveness of classical island models largely depends on these migration mechanisms, which influence the balance between exploration, exploitation and information dissemination among subpopulations. Motivated by these limitations, this work proposes a novel cooperative parallel framework termed Parallel Enzyme Action Optimizer (PEAO), which extends the original EAO through distributed population evolution and structured information exchange mechanisms. Although cooperative island-based evolutionary algorithms and distributed population models have been extensively investigated in the literature, the proposed PEAO should not be interpreted as a conventional island-model implementation. Conventional island models rely primarily on predefined migration policies that determine

when, how many and which individuals are exchanged among islands. Rather than relying solely on predefined migration policies, PEAO extends the original EAO by integrating adaptive propagation, multiple communication strategies, optional local search refinement and a convergence-based stopping mechanism within a unified cooperative optimization framework. Consequently, the proposed method combines several complementary mechanisms that jointly improve information dissemination, preserve population diversity and enhance the balance between exploration and exploitation, rather than simply parallelizing the original algorithm. Instead of evolving a single population, the search process is decomposed into several semi-independent subpopulations that perform exploration simultaneously while exchanging information through the proposed adaptive communication and propagation mechanisms. A key component of the proposed framework is the introduction of an adaptive propagation mechanism that enables the controlled dissemination of high-quality solutions among clusters. By facilitating the transfer of promising search information across different regions of the population, the propagation process accelerates convergence while preserving sufficient diversity to maintain exploration capabilities. Although a recent study introduced a parallel EAO-based optimizer incorporating a multi-strategy communication mechanism for breast cancer diagnosis [21], its primary objective was hyperparameter optimization for a specific machine learning application. In contrast, the proposed framework is developed as a general-purpose continuous optimization algorithm and further extends the EAO framework through the integration of a K-means-based population initialization, adaptive propagation, optional local search, a convergence-based stopping mechanism and a systematic investigation of communication strategies.

The main contributions of the proposed framework are summarized as follows:

- A cooperative parallel extension of the Enzyme Action Optimizer is proposed, where the initial population is partitioned into cooperative subpopulations using K-means clustering prior to parallel optimization.
- A propagation mechanism is introduced to facilitate the exchange of high-quality solutions among subpopulations, accelerating convergence while maintaining population diversity.
- Multiple communication strategies are incorporated, enabling diverse forms of information exchange and cooperative interaction among subpopulations.
- A convergence-based stopping criterion that monitors successive improvements of the best objective function value.
- A scalable optimization framework is developed that improves the balance between exploration and exploitation and is particularly suitable for high-dimensional and computationally demanding optimization problems.

Unlike conventional island models, where cooperation is typically achieved through predefined migration policies, the proposed PEAO integrates adaptive propagation, multiple communication strategies, convergence-based stopping and optional local refinement into a unified cooperative optimization framework. Furthermore, unlike previous parallel EAO variants developed for specific application domains, the proposed method is designed as a general-purpose optimizer for continuous black-box optimization problems. Consequently, the proposed framework extends the original EAO through the integration of multiple complementary search mechanisms rather than by simply parallelizing the original algorithm. The proposed PEAO is intended for continuous, single-objective, bound-constrained optimization problems, particularly black-box problems for which analytical gradient information is unavailable to the global search mechanism. Therefore, PEAO should be regarded as an extension of the original EAO rather than a direct parallel implementation, since it combines multiple complementary algorithmic mechanisms that fundamentally

modify both the search dynamics and the cooperation among individuals. The method primarily targets non-linear, multimodal and high-dimensional problems in which maintaining an effective balance between exploration and exploitation is essential. The core population-based search process remains gradient-free, while the optional BFGS procedure is employed only as a local refinement mechanism when applicable.

The remainder of this paper is organized as follows. Section 2 describes the proposed Parallel Enzyme Action Optimizer (PEAO) framework and its fundamental components. Section 3 presents the experimental setup, benchmark problems, and evaluation methodology. Section 4 reports and discusses the experimental results, including an analysis of the contributions of the proposed algorithmic components. Finally, section 5 concludes the paper and outlines directions for future research.

2. Materials and Methods

2.1. PEAO Algorithm

The proposed Parallel Enzyme Action Optimizer (PEAO) is a population-based meta-heuristic designed for continuous, single-objective, bound-constrained optimization problems. Its global population-based search mechanism does not require analytical derivative information and is therefore applicable to continuous black-box optimization problems. PEAO aims to maintain an effective balance between global exploration and local exploitation while enhancing population diversity through parallel cooperative search. In PEAO, the population is divided into multiple semi-independent subpopulations that evolve concurrently within the D -dimensional search space and periodically cooperate through the proposed communication and propagation mechanisms. Each individual x_i represents a candidate solution to the optimization problem. Initially, a population of N individuals is randomly generated within the prescribed search bounds. The generated population is subsequently partitioned into K cooperative subpopulations using the K-means clustering algorithm, where each cluster forms one subpopulation for the subsequent parallel optimization process. This K-means procedure is applied only once during initialization to group similar candidate solutions into cooperative subpopulations and provide a structured decomposition of the initial population. The objective function is evaluated for each individual and the best solution identified so far is denoted by x_{best} . The structure of PEAO was selected so that each algorithmic component addresses a distinct requirement of complex continuous optimization problems. The K-means-based initialization organizes the initial population into cooperative subpopulations, facilitating the simultaneous exploration of different regions of the search space. The sinusoidal operator introduces non-linear perturbations around promising solutions, whereas the Lévy-flight and population-difference operators generate long-range and population-guided search movements. The greedy selection mechanism preserves improvements, the optional local search procedure refines promising solutions and the convergence-based stopping mechanism limits unnecessary objective function evaluations. Therefore, PEAO combines complementary mechanisms intended to address exploration, exploitation, diversity preservation, local refinement and computational efficiency rather than relying on a single update operator. During the optimization process, an adaptive factor A is dynamically adjusted according to $A = \sqrt{\frac{\text{iter}}{T}}$, where $\text{iter} \in 0, \dots, T$ denotes the current iteration and T is the maximum number of iterations. The adaptive factor increases monotonically from 0 to 1 and directly controls the contribution of the attraction term toward the current best solution in the candidate-generation equations. Compared with a linear schedule, the square-root formulation increases more rapidly during the early and intermediate stages of the optimization process and more gradually near its final stage. Consequently, guidance from the best solution is incorporated relatively early, while the sinusoidal, Lévy-flight

and population-difference components continue to provide stochastic exploration. The selected schedule therefore combines progressive best-solution guidance with persistent stochastic perturbations throughout the search process. For each individual, three candidate solutions are generated using complementary search mechanisms. The first candidate is generated through a sinusoidal perturbation strategy guided by the current best solution. This operator introduces non-linear oscillatory movements around promising regions of the search space and increases the diversity of the generated candidates. The sinusoidal component allows the direction and magnitude of the perturbation to vary during the optimization process, thereby reducing the probability that all individuals follow identical search trajectories. The second and third candidate-generation mechanisms combine Lévy-flight perturbations, population-difference information and attraction toward the current best solution. Two randomly selected individuals are first used to construct a difference vector, which introduces information about the spatial distribution of the current population. The Lévy-flight component additionally provides stochastic movements with occasional long steps, enabling the algorithm to investigate regions located away from the current population. Although the two candidates, denoted by *C2A* and *C2B*, employ the same general sources of search information, they were designed to generate complementary search patterns. In *C2A*, both the Lévy perturbation and the scaling factor are vectors. Consequently, each decision variable may receive a different perturbation coefficient, resulting in dimension-specific and anisotropic search movements. This mechanism increases the diversity of the generated solutions and supports the investigation of different directions within the search space. In contrast, *C2B* employs a scalar Lévy coefficient and a scalar scaling factor. The same scalar coefficient is therefore applied across all decision variables, producing more coordinated movements along the population-difference and best-solution attraction directions. The coexistence of vector-based and scalar-based mechanisms provides complementary dimension-wise and coordinated search behaviors. Their combined use increases the variety of candidate solutions generated during each iteration and reduces dependence on a single search pattern. After the three candidate solutions have been generated, their objective function values are evaluated and the best candidate is selected. A greedy selection mechanism is subsequently applied. For a minimization problem, the current solution is replaced only when the selected candidate has an objective function value equal to or lower than that of the current solution. Consequently, accepted updates do not deteriorate the fitness of the corresponding individual. To further enhance local refinement, a local search procedure may be activated with probability $p_l = 0.05$. When triggered, the BFGS algorithm is applied to a promising current solution as an optional refinement stage. Since this procedure is invoked with a low probability, it does not constitute the primary search component of PEO. The population-based candidate-generation process remains derivative-free, whereas BFGS is employed only when the objective function is sufficiently smooth and gradient information can be supplied or numerically approximated. For non-differentiable problems, the local-search stage can be disabled by setting $p_l = 0$ or replaced by an appropriate derivative-free local optimizer. Therefore, the global search mechanism of PEO remains applicable to continuous black-box optimization, while the optional BFGS stage provides additional local refinement when appropriate. The BFGS local search is executed using the default stopping criteria of the optimization library implementation. Since local refinement is activated with a probability of only 5%, it does not dominate the computational cost of the proposed method. Furthermore, all objective function evaluations performed during the local search phase are included in the reported total number of objective function evaluations. At the end of each iteration, the global best solution is updated using the objective function values of the newly evaluated solu-

Algorithm 1 The PEO algorithm**INPUT**

- f : objective function
- N : population size
- D : problem dimension
- T : maximum number of iterations
- EC : enzyme concentration
- β : Levy parameter
- N_t : termination criterion
- $p_l \in [0, 1]$ as the local search rate.

OUTPUT

- x_{best} : best solution

INITIALIZATION

- **Set** $iter \leftarrow 0$ as the iteration counter
- **Initialize individuals** $x_i \in R^D, i \in \{1..N\}$
- **Compute** fitness $F_i \leftarrow f(x_i)$ for each individual
- **Identify** best solution: x_{best}

main pseudocode

```

01 while stopping criterion is not met do
02   Compute adaptive factor:  $A \leftarrow \sqrt{\frac{iter}{T}}$ 
03   for each individual  $i, i \in \{1..N\}$  do
04     Draw  $r \sim U(0, 1)$ 
05     Generate Candidate 1:  $C_1 \leftarrow (x_{best} - x_i) + r \cdot \sin(A \cdot x_i)$ 
06     Select two random individuals  $x_p$  and  $x_q$ 
07     Compute difference vector:  $DV \leftarrow x_p - x_q$ 
08     Generate Levy vector  $Lu_D(\beta) \leftarrow Levy(D, \beta)$ 
09     Generate Candidate 2A:  $C2A \leftarrow x_i + Lu_D(\beta) \cdot DV + A \cdot scA \cdot (x_{best} - x_i)$  where  $scA \in [EC, 1]^D$ 
10     Generate Levy scalar  $Lu_1(\beta) \leftarrow Levy(1, \beta)$ 
11     Generate Candidate 2B:  $C2B \leftarrow x_i + Lu_1(\beta) \cdot DV + A \cdot scB \cdot (x_{best} - x_i)$  where  $scB \in [EC, 1]$ 
12     Evaluate  $C_1, C2A, C2B$ 
13     Select the best candidate:  $C \leftarrow \operatorname{argmin}(f(C_1), f(C2A), f(C2B))$ 
14     if  $f(C) \leq f(x_i)$  then
15        $x_i \leftarrow C$ 
16        $F_i \leftarrow f(C)$ 
17     Endif
18     Draw  $r \sim U(0, 1)$ 
19     if  $r \leq p_l$  then
20       Apply local search  $x_i \leftarrow LS(x_i)[22]$ 
21        $F_i \leftarrow f(x_i)$ 
22     end if
23   endfor
24   Set  $iter = iter + 1$ 
25   if  $iter \geq T$  then terminate
26     Evaluate  $F_i \leftarrow f(x_i)$  for all individuals
27     Update  $x_{best}$  based on best fitness value
28     Compute  $\delta^{(iter)} = |f_{min}^{(iter)} - f_{min}^{(iter-1)}| [23]$ 
29     if  $\delta^{(iter)} \leq \epsilon$  for  $N_t$  iterations then terminate.
30     Update  $x_{best}$  based on best fitness value
31 end while // Goto termination check
Return  $x_{best}$ 

```

tions. The convergence-based stopping criterion is then applied by monitoring the absolute change in the best objective function value between two consecutive iterations:

$\delta^{(iter)} = |f_{min}^{(iter)} - f_{min}^{(iter-1)}|$. If $\delta^{(iter)}$ remains below a predefined tolerance ϵ for N_t consecutive iterations, the optimization process is considered to have converged and is terminated. In this context, the term Similarity refers to the similarity of the optimization progress observed in successive iterations rather than to similarity among individuals, clusters or subpopulations. This stopping mechanism reduces unnecessary objective function evaluations when no meaningful improvement is detected. By combining sinusoidal perturbation, Lévy-flight movements, population-difference learning, adaptive best-solution guidance, greedy selection, optional local refinement and convergence-based termination, PEO provides multiple complementary search behaviors within a unified optimization framework. These mechanisms are intended to maintain population diversity, reduce premature convergence, refine promising solutions and limit unnecessary objective function evaluations when solving complex continuous optimization problems.

2.2. Proposed PEO Framework

Figure 2 presents the parallel execution architecture and communication mechanisms of the proposed PEO framework. Before the parallel optimization process begins, the initial population is partitioned into K cooperative subpopulations using the K-means

clustering procedure. Each resulting cluster is assigned to a dedicated processing unit (PU). Each Processing Unit (PU) is responsible for the evolution of a single subpopulation within which the core PEO search procedure is executed independently. Candidate solutions are generated using the C1, C2A and C2B update mechanisms, while the most promising candidate is selected through a greedy replacement strategy. The subpopulations evolve simultaneously and independently, enabling the parallel exploration of different regions of the search space and promoting population diversity. After each search cycle, the overall best solution is updated and the communication conditions among subpopulations are evaluated. When communication is activated, one of the available information-exchange strategies is applied, namely Optimal Replacement, Optimal Mean Replacement or Circular Optimal Replacement. These strategies facilitate the transfer of useful search knowledge among subpopulations by replacing low-quality solutions with high-quality information obtained from other search groups. In addition, the propagation mechanism supports the controlled dissemination of promising solutions across processing units, enabling effective cooperation among subpopulations while preserving their exploration capabilities. The combination of parallel exploration, controlled information propagation and multi-strategy communication contributes to maintaining population diversity, accelerating convergence and strengthening the balance between exploration and exploitation. As a result, the proposed PEO framework achieves improved search efficiency and robustness when solving complex optimization problems. Furthermore, the steps of the proposed method are also graphically shown in Figure 1.

Algorithm 2 Proposed PEO Framework

INPUT
 – f : objective function
 – K : number of subpopulations
 – C_R : communication threshold
 – C_S : communication strategy mode
 – $C_S \in \{0, 1, 2, 3\}$
 – N_M : propagation method
 – N_R : propagation rate
 – N_P : number of propagation agents
 – N_I : termination criterion
OUTPUT
 (x_{best}) : global best solution
INITIALIZATION
 Set $t \leftarrow 0$
 Initialize population
 Partition population into K cooperative subpopulations using K-means clustering
 Compute fitness values
 Identify local best and worst individuals
 Identify global best solution x_{best}
MAIN PSEUDOCODE
 01 **while** stopping criterion is not met **do**
 02 Set $t \leftarrow t + 1$
 03 **for** each subpopulation $k \in 1..K$ in parallel **do**
 04 Execute one iteration of the Algorithm 1
 05 Update local best and worst individuals
 06 **if** $N_R > 0$ and $t \bmod N_R = 0$ **then**
 07 Apply propagation scheme N_M with N_P agents
 08 **endif**
 09 **endfor**
 10 Update global best solution x_{best}
 11 **if** $C_R > 0$ and $t \bmod C_R = 0$ **then**
 12 **if** $C_S = 0$ **then**
 13 $c \leftarrow$ random integer $\in \{1, 2, 3\}$
 14 **else**
 15 $c \leftarrow C_S$
 16 **endif**
 17 **if** $c = 1$ **then**
 18 Apply Optimal Replacement Strategy
 19 **elseif** $c = 2$ **then**
 20 Apply Optimal Mean Replacement Strategy
 21 **else**
 22 Apply Circular Optimal Replacement Strategy
 23 **endif**
 24 Update local best and worst individuals
 25 Update global best solution x_{best}
 26 **end while**
 Return x_{best}

2.3. Propagation Mechanism

The propagation mechanism facilitates the exchange of information among subpopulations by periodically sharing high-quality candidate solutions throughout the optimization process. During each propagation stage, the best solution identified within a selected subpopulation is transferred to one or more target subpopulations, where it replaces the poorest-performing individual. By continuously introducing promising solutions across subpopulations, this mechanism strengthens cooperation, promotes knowledge sharing and guides the search toward more promising regions of the solution space. To ensure low communication overhead while preserving scalability, the proposed framework employs lightweight propagation schemes rather than conventional migration policies. The following propagation strategies are investigated:

- One-to-One (1to1): A randomly chosen subpopulation transfers its current best solution to another randomly selected subpopulation.
- One-to-All (1toN): The best solution found by a randomly selected subpopulation is distributed to every remaining subpopulation.
- All-to-One (Nto1): The best solutions produced by all subpopulations are transmitted to a single randomly selected recipient subpopulation.
- All-to-All (NtoN): Every subpopulation shares its best solution with each of the remaining subpopulations, enabling comprehensive information exchange.

Collectively, these propagation strategies facilitate the dissemination of high-quality candidate solutions while maintaining the computational efficiency and scalability of the proposed parallel optimization framework.

2.4. Communication Strategies

2.4.1. Optimal Replacement

During communication, the best individuals from all cooperating subpopulations are propagated to each subpopulation, where they replace the worst individual. This strategy enables the rapid dissemination of high-quality search information while preserving the remaining individuals, thereby accelerating convergence and improving the overall search efficiency.

2.4.2. Optimal Mean Replacement

During communication, a new candidate solution is constructed as the arithmetic mean of the best individuals from all cooperating subpopulations. The generated solution is subsequently used to replace the worst individual within each destination subpopulation. By integrating information from multiple high-quality solutions, this strategy facilitates smoother knowledge sharing, improves population diversity and promotes a more balanced trade-off between exploration and exploitation.

2.4.3. Circular Optimal Replacement

The subpopulations are organized in a logical ring topology. During each communication phase, the best individual from the preceding subpopulation is transferred to the subsequent subpopulation, where it replaces the worst individual. Consequently, high-quality search information is propagated progressively throughout the population, promoting balanced cooperation among subpopulations while preserving diversity.

3. Experiments

This section describes the experimental methodology and presents the performance evaluation of the proposed PEA algorithm. It begins with the benchmark functions, experimental setup, and parameter configuration. The impact of the main algorithmic com-

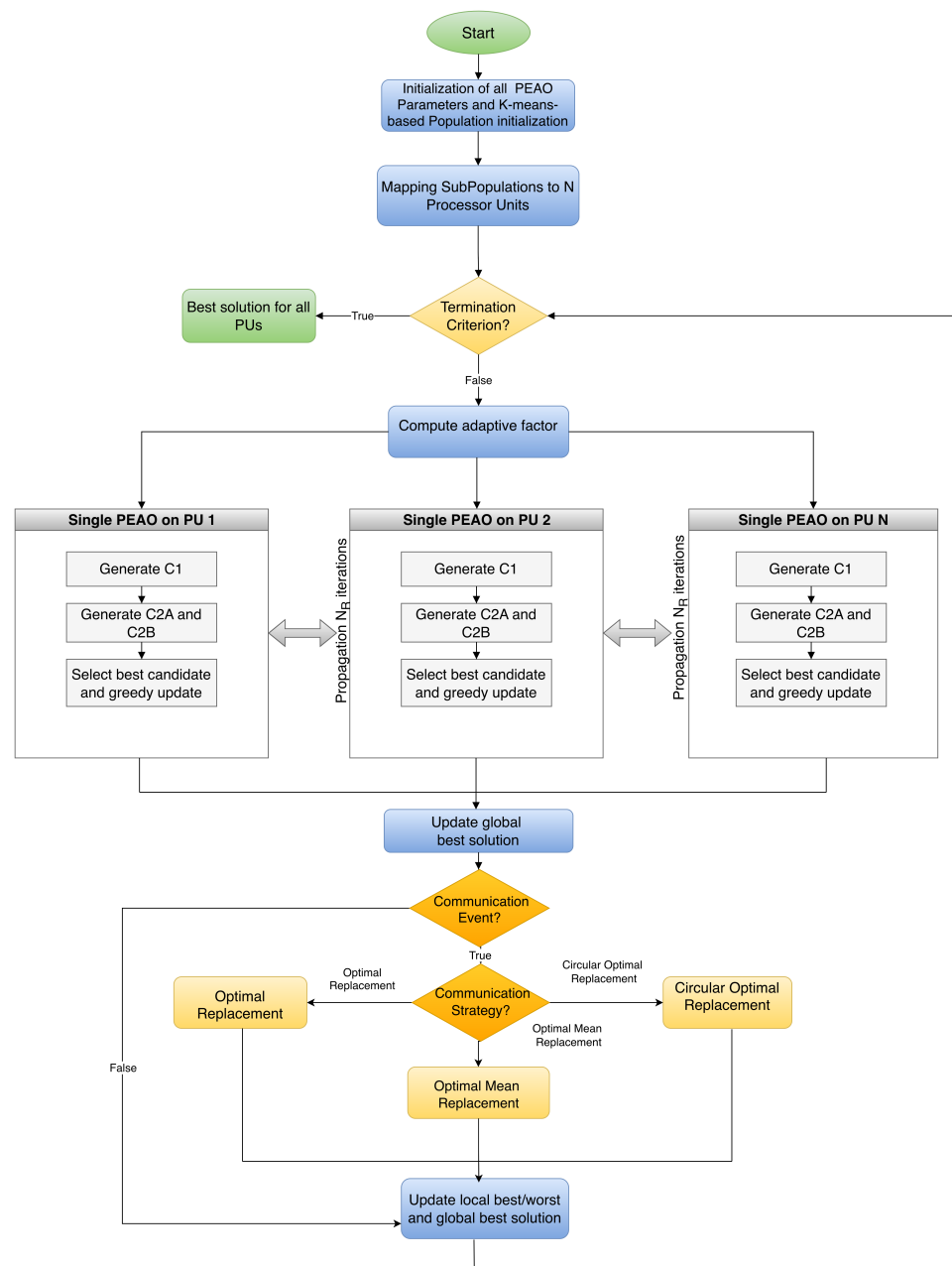


Figure 1. Flowchart of the proposed PEAO framework.

Randomized Communication Strategies among Subpopulations in PEO

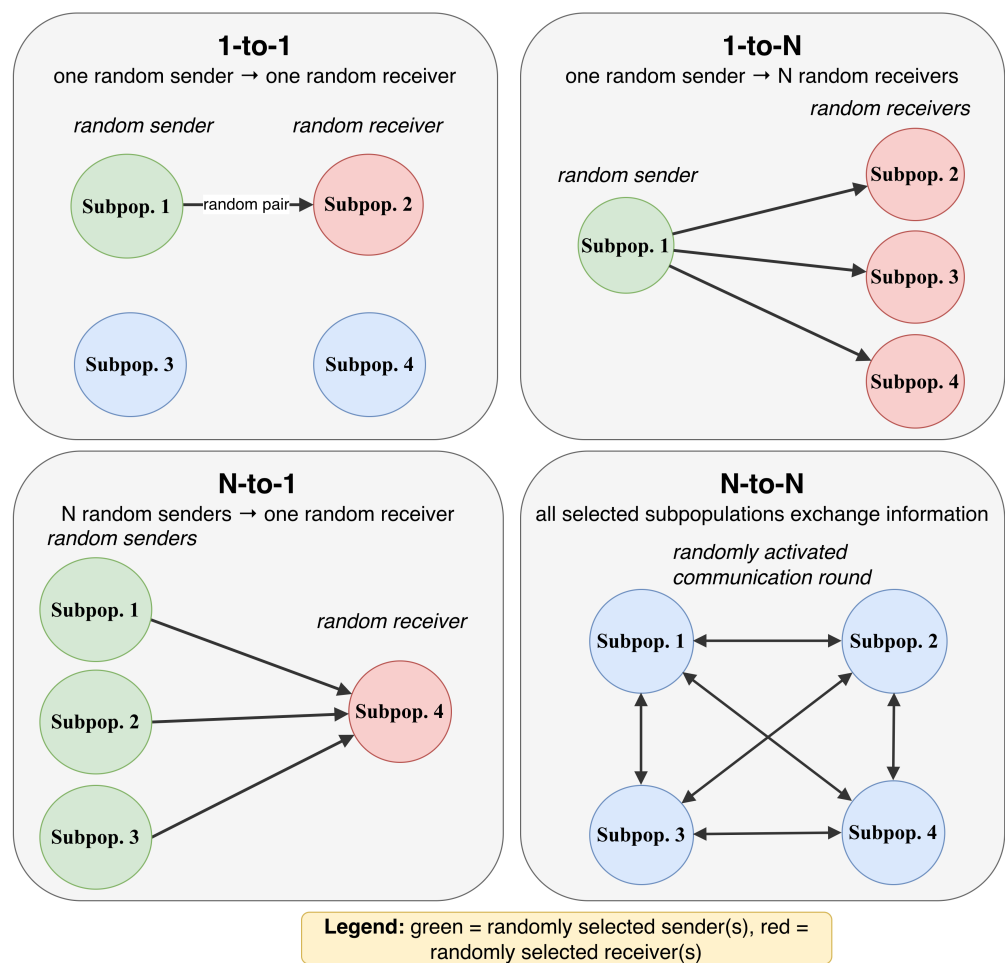


Figure 2. Communication and propagation strategies among subpopulations.

ponents, including the number of subpopulations, propagation strategies, communication mechanisms, adaptive search factor and parallel implementation, is then systematically investigated. Finally, comparative experiments with representative optimization algorithms are presented to evaluate the effectiveness, robustness and computational efficiency of the proposed method.

3.1. Experiments Setup

3.1.1. Benchmark Functions

A diverse collection of established continuous optimization benchmark functions was selected to evaluate the proposed PEO algorithm. Several of these functions constitute the fundamental building blocks of widely adopted benchmark suites, including BBOB and CEC, where they are typically combined to construct transformed and composite benchmark problems. In this work, the original formulations of these functions were considered in order to provide a direct and transparent assessment of the proposed method over a wide range of optimization landscape characteristics. [24–27]. The selected benchmark set covers a variety of problem characteristics, including unimodal, multimodal, ill-conditioned, weak global structure and non-separable functions. Consequently, the experimental evaluation examines the algorithm’s convergence behavior, exploration and exploitation capabilities, robustness against local optima, and scalability under different optimization scenarios. All benchmark problems were tested using dimensionalities ranging from 25 to 150 variables, where n denotes the problem dimension. Table 1 summarizes the benchmark functions considered in this study together with their mathematical definitions, problem dimensionalities, search intervals and corresponding global optimum values.

Table 1. Benchmark test functions used in experimental study.

NAME	FORMULA	DIM	BOUNDS	G_{min}
ATTRACTIVE SECTOR	$f(x) = \left(\sum_{i=1}^n (s_i x_i)^2\right)^{0.9}$	n	$[-5, 5]^n$	0
BÜCHE RASTRIGIN	$f(x) = \sum_{i=1}^n \left[z_i \cdot (1 + 0.1 \cdot \sin(10\pi z_i)) \right]$	n	$[-5.12, 5.12]^n$	0
DIFFERENT POWERS	$f(x) = \sqrt{\sum_{i=1}^n x_i ^{2+4\frac{i-1}{n-1}}}$	n	$[-5, 5]^n$	0
DISCUS	$f(x) = 10^6 x_1^2 + \sum_{i=2}^n x_i^2$	n	$[-5, 5]^n$	0
ELLIPSOIDAL	$f(x) = \sum_{i=1}^n \left(10^6\right)^{\frac{i-1}{n-1}} x_i^2$	n	$[-5.12, 5.12]^n$	0
GALLAGHER101	$f(x) = \max_{i=1}^{101} \left[h_i - w_i \sqrt{\sum_{j=1}^n (x_j - c_{ij})^2} \right] \min = 101$	n	$[-5, 5]^n$	0
GALLAGHER21	$f(x) = \max_{i=1}^{21} \left[h_i - w_i \sqrt{\sum_{j=1}^n (x_j - c_{ij})^2} \right] \min = 21$	n	$[-5, 5]^n$	0
GRIEWANK ROSENBROCK	$f(x) = \left(\frac{\ x\ ^2}{4000} - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1 \right) \cdot \left(\frac{1}{10} \sum_{i=1}^{n-1} \left[100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2 \right] \right)$	n	$[-5, 5]^n$	0
GRIEWANK	$f(x) = 1 + \frac{1}{200} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \frac{\cos(x_i)}{\sqrt{(i)}}$	n	$[-100, 100]^n$	0
RASTRIGIN	$f(x) = An + \sum_{i=1}^n \left[x_i^2 - A \cos(2\pi x_i) \right] A = 10$	n	$[-5.12, 5.12]^n$	0
ROSENBROCK	$f(x) = \sum_{i=1}^{n-1} \left(100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2 \right), \quad -30 \leq x_i \leq 30$	n	$[-30, 30]^n$	0
ROTATED ROSENBROCK	$f(x) = \sum_{i=1}^{n-1} \left[100(z_{i+1} - z_i^2)^2 + (z_i - 1)^2 \right], \quad z = Rx$	n	$[-5, 5]^n$	0
KATSUURA	$f(x) = \frac{10}{n^2} \prod_{i=1}^n \left(1 + i \sum_{k=1}^{32} \frac{ 2^k x_i - \lfloor 2^k x_i \rfloor }{2^k} \right)^{\frac{10}{n^2-2}} - \frac{10}{n^2} + \frac{1}{D^n} \sum_{i=1}^n x_i^2$	n	$[0.0, 1.0]^n$	0
SHARP RIDGE	$f(x) = x_1^2 + a \sum_{i=2}^n x_i^2, \quad a > 1$	n	$[-5, 5]^n$	0
STEP ELLIPSOIDAL	$f(x) = \sum_{i=1}^n [x_i + 0.5]^2 + a \sum_{i=1}^n \left(10^6 \cdot \frac{i-1}{n-1} \right) x_i^2, \quad a = 1$	n	$[-5, 5]^n$	0
WEIERSTRASS	$f(x) = \sum_{i=1}^n \left[\sum_{k=0}^{k_{\max}} \left(a^k \cos(2\pi b^k (x_i + 0.5)) \right) \right] - n \sum_{k=0}^{k_{\max}} \left(a^k \cos(2\pi b^k \cdot 0.5) \right)$ $a = 0.5, b = 3, k_{\max} = 20$	n	$[-0.5, 0.5]^n$	0
ZAKHAROV	$f(x) = \sum_{i=1}^n x_i^2 + \left(\sum_{i=1}^{n-1} \frac{1}{i} x_i \right)^2 + \left(\sum_{i=1}^{n-1} \frac{1}{i} x_i \right)^4$	n	$[-5, 10]^n$	0

Table 2. Characteristics of the benchmark functions.

NAME	CATEGORY	EVALUATED PROPERTY
ATTRACTIVE SECTOR	Ill-conditioned	Local exploitation capability
BUCHER RASTRIGIN	Multimodal	Escaping local optima
DIFFERENT POWERS	Unimodal	Convergence speed
DISCUS	Ill-conditioned	Numerical stability
ELLIPSOIDAL	Ill-conditioned	Performance on ill-conditioned problems
GALLAGHER101	Highly Multimodal	Population diversity
GALLAGHER21	Highly Multimodal	Global exploration capability
GRIEWANK ROSENBROCK	Composite	Hybrid search capability
GRIEWANK	Multimodal	Exploration-exploitation balance
RASTRIGIN	Multimodal	Avoiding premature convergence
ROSENBROCK	Valley-shaped	Local search capability
ROTATED ROSENBROCK	Non-separable	Variable interaction handling
KATSUURA	Rugged	Search robustness
SHARP RIDGE	Ridge	Exploration capability
STEP ELLIPSOIDAL	Discontinuous	Performance on discontinuous landscapes
WEIERSTRASS	Fractal	Performance on rugged landscapes
ZAKHAROV	Unimodal	Convergence speed

The benchmark suite was intentionally selected to cover a broad spectrum of optimization challenges. Unimodal functions mainly evaluate convergence speed and exploitation capability, whereas multimodal functions assess the ability of the optimizer to avoid premature convergence and escape local optima. Ill-conditioned functions examine numerical robustness under narrow search valleys, while composite and weak global structure functions evaluate the balance between exploration and exploitation. Finally, non-separable problems assess the capability of the algorithm to optimize strongly interacting decision variables, which are frequently encountered in real-world optimization problems.

3.1.2. Experimental Environment

All benchmark functions described in the previous subsection were evaluated under the same experimental conditions. The experiments were performed on a workstation equipped with an AMD RYZEN 5950X processor and 128 GB of RAM, running the Debian Linux operating system. To ensure statistical reliability, each benchmark problem was independently executed 30 times, using a different random seed for every run, and the reported results correspond to the average performance over these independent executions. The proposed algorithm was implemented in C++ within the publicly available OPTIMUS optimization framework [28] which can be downloaded from <https://github.com/itsoulos/OPTIMUS> (accessed on 23 June 2026). Employing an open-source optimization framework enhances the reproducibility of the experimental study and enables independent verification of the reported results by other researchers.

3.1.3. Parameter Setting and Evaluation Protocol

All optimization algorithms included in the comparative study were implemented independently within the same experimental framework rather than directly using existing implementations from the literature. To ensure a fair and consistent comparison, every algorithm was developed in C++ and executed under identical hardware and software conditions. In addition, the same stopping criterion, population size, maximum number of iterations, and evaluation protocol were adopted for all methods, while identical parameter values were used whenever applicable. For all algorithms, the population size was fixed at 200 individuals, the maximum number of iterations was set to 200 and the local search probability was maintained at $p_l = 0.05$. Using a common experimental configuration ensures that the observed performance differences primarily reflect the characteristics of the optimization algorithms rather than differences in the evaluation environment. Algorithm-specific parameters were selected according to the recommendations reported in

the corresponding publications and the original EAO algorithm, and these values were kept unchanged throughout all benchmark problems without problem-dependent parameter tuning. Although a dedicated sensitivity analysis of all parameters is beyond the scope of this work, the proposed method demonstrated stable and consistent performance under the selected configuration. The complete parameter settings employed in the experiments are summarized in Table 3.

Table 3. The values for the experimental parameters.

PARAMETER	VALUE	EXPLANATION
N	200	Number of individuals
$iter_{max}$	200	Maximum number of iterations
K	1,3,10,20,30	Number of subpopulations
SR	$\delta_k^{(iter)} = f_{k,min}^{(iter)} - f_{k,min}^{(iter-1)} $	Stopping rule:
P_L	0.05 (5%)	Local search rate
N_M	1to1, 1toN, Nto1, NtoN	Propagation method
N_R	$1,2,3,4 \dots \leq N$	Number of iterations for propagation
N_P	$1,2,3,4 \dots \leq N$	Number of agents for propagation
N_T	$1,2,3,4 \dots \leq N$	Number of subpopulations required to satisfy the stopping criterion
F	0.8	Differential weight for DE
CR	0.9	Crossover probability
N_I	8	Number of iterations used in the termination rule
-	0.05 (5%)	Mutation rate for GA
-	0.05 (5%)	Selection Rate for GA
-	Roulette	Selection method for GA

At iteration $iter$, the convergence-based stopping criterion computes the absolute difference between the best objective-function values obtained in two consecutive iterations: $\delta^{(iter)} = |f_{min}^{(iter)} - f_{min}^{(iter-1)}|$. A run terminates when $\delta^{(iter)} \leq \varepsilon$ for $N_I = 8$ consecutive iterations or when the maximum number of iterations is reached. In all experiments, the tolerance was set to $\varepsilon = 10^{-6}$. Termination indicates that no meaningful improvement has been observed according to the adopted stopping criterion and does not necessarily imply that the global optimum has been identified. The reported number of objective function evaluations corresponds to the total number of objective function calls performed from initialization until the stopping criterion is satisfied. Objective function evaluations incurred during candidate generation and local search refinement were also included in the reported evaluation count. This evaluation protocol is consistent with common practice in benchmarking evolutionary optimization algorithms, where all objective function evaluations contribute to the computational effort. For benchmark functions with a known global optimum f^* , the same numerical tolerance was adopted to determine whether a run was successful. Specifically, a run was considered successful if $|f_{min} - f^*| \leq \varepsilon$. Runs that terminated without satisfying this condition were not discarded instead, their final objective values and the corresponding numbers of objective function evaluations were included in the reported statistics. Consequently, the reported evaluation counts represent the computational effort consumed until termination rather than the number of evaluations required to reach the global optimum. Values reported in parentheses indicate the success rate, expressed as the proportion of successful runs among the 30 independent executions.

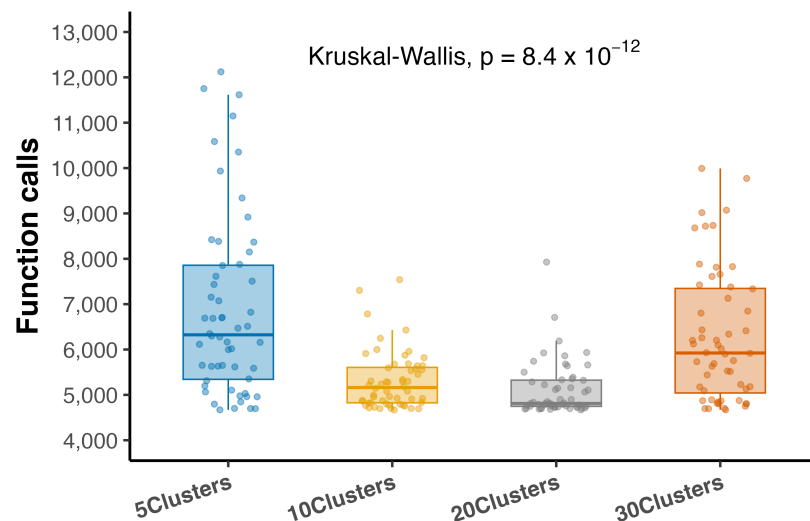


Figure 3. Effect of the Number of Clusters on Optimization Performance.

3.2. Sensitivity Analysis of PEA0

3.2.1. Effect of the Number of Subpopulations

Table 4 compares the average number of objective function evaluations obtained using different numbers of subpopulations (5, 10, 20 and 30). Lower values indicate better computational efficiency. Overall, the number of subpopulations has a substantial impact on the performance of the proposed PEA0 algorithm. The configurations with 10 and 20 subpopulations consistently achieve the lowest computational effort across the considered benchmark problems, whereas 5 and 30 subpopulations generally require a larger number of objective function evaluations. The cumulative results further support this observation, with the total number of objective function evaluations decreasing from 385,708 for 5 subpopulations to 297,299 and 286,096 for 10 and 20 subpopulations, respectively, before increasing to 351,709 for 30 subpopulations. These findings indicate that a moderate number of subpopulations provides the best balance between exploration and exploitation, resulting in improved convergence and computational efficiency under the examined experimental configuration.

Table 4. Comparison of average objective function evaluations for different numbers of subpopulations.

FUNCTION	5 CLUSTER	10 CLUSTER	20 CLUSTER	30 CLUSTER
BUCHER RASTRIGIN_25	6298	4866	4807	5917
BUCHER RASTRIGIN_50	8368	5874	5679	7336
BUCHER RASTRIGIN_100	10352	6429	6188	9071
BUCHER RASTRIGIN_150	10585	4715	5934	9771
SHARP RIDGE_25	6158	4795	4713	6099
SHARP RIDGE_50	6707	5338	4835	5898
SHARP RIDGE_100	7153	5644	4793	6017
SHARP RIDGE_150	6699	5640	4850	5930
ZAKHAROV_25	5312	4827	4721	5092
ZAKHAROV_50	6350	5179	4770	5756
ZAKHAROV_100	8384	4803	4718	6847
ZAKHAROV_150	7434	4919	4896	7827

Figure 3 presents the influence of the number of clusters on the computational efficiency of the proposed algorithm. The results reveal that the configurations employing 10 and 20 clusters require significantly fewer function evaluations than the alternatives while also demonstrating lower variability across independent runs. Conversely, the 5-cluster

configuration exhibits the highest median number of function calls and substantial dispersion, indicating slower and less reliable convergence. Although the 30-cluster configuration performs better than the 5-cluster case, its variability remains considerably higher than that observed for 10 and 20 clusters. The Kruskal–Wallis test ($p = 8.4 \times 10^{-12}$) indicates statistically significant differences among the examined configurations. These findings suggest that using a moderate number of clusters can substantially enhance the search efficiency and robustness of the algorithm, with 10 and 20 clusters yielding the most favorable performance.

3.2.2. Effect of the Propagation Mechanism

Table 5 compares the proposed PEAO algorithm with and without the propagation mechanism (PROP = 1 and PROP = 0) in terms of the average number of objective function evaluations. Lower values indicate better computational efficiency. Overall, enabling the propagation mechanism consistently reduces the number of objective function evaluations across all benchmark problems, with the most pronounced improvements observed for the higher-dimensional benchmark instances. The cumulative results further demonstrate the effectiveness of propagation, reducing the total number of objective function evaluations from 1,897,933 without propagation to 297,299 with propagation, corresponding to an overall reduction of approximately 84%. These results indicate that the propagation mechanism substantially enhances information sharing among subpopulations, accelerates convergence, and significantly improves the computational efficiency of the proposed framework.

Table 5. Effect of the proposed propagation mechanism on average objective function evaluations.

FUNCTION	PROP=0	PROP=1
BUCHE RASTRIGIN_25	58030	4866
BUCHE RASTRIGIN_50	86188	5874
BUCHE RASTRIGIN_100	116211	6429
BUCHE RASTRIGIN_150	135523	4715
SHARP RIDGE_25	18646	4795
SHARP RIDGE_50	19932	5338
SHARP RIDGE_100	20959	5644
SHARP RIDGE_150	21109	5640
ZAKHAROV_25	10499	4827
ZAKHAROV_50	15031	5179
ZAKHAROV_100	23334	4803
ZAKHAROV_150	28647	4919

Figure 4 illustrates the effect of the propagation mechanism on the performance of the proposed framework. PROP = 1 denotes the configuration with propagation enabled, whereas PROP = 0 corresponds to the configuration without propagation. Enabling propagation consistently reduces the number of objective function evaluations and results in a more concentrated distribution of the obtained values. A paired Wilcoxon signed-rank test confirmed that the difference between the two configurations is statistically significant ($p < 0.0001$). Statistical significance is indicated using the conventional notation (ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$). These results indicate that the propagation mechanism improves the convergence behavior of the proposed framework under the considered experimental conditions.

3.2.3. Effect of Propagation Methods with 10 Subpopulations

Table 6 compares the four propagation strategies (1to1, 1toN, Nto1 and NtoN) using a fixed configuration of 10 subpopulations. Lower numbers of objective function evaluations indicate better optimization performance. Overall, the NtoN communication strategy

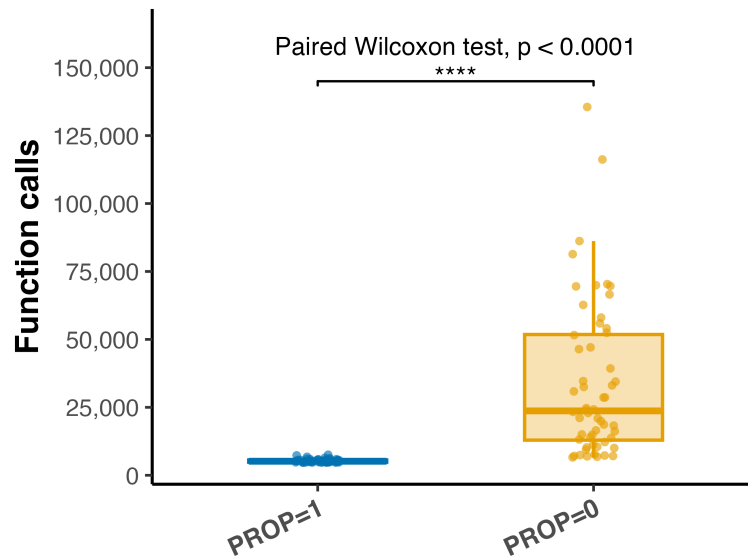


Figure 4. Impact of the Propagation Mechanism.

consistently achieves the lowest computational effort across the considered benchmark problems, whereas the 1to1 strategy generally requires the highest number of objective function evaluations. The 1toN and Nto1 strategies provide intermediate performance. The cumulative results further confirm the superiority of the NtoN strategy, reducing the total number of objective function evaluations from 982,812 for the 1to1 strategy to 297,299, corresponding to reductions of approximately 70%, 33% and 43% compared with the 1to1, 1toN, and Nto1 strategies, respectively. These findings indicate that simultaneous information exchange among all subpopulations substantially improves cooperation and accelerates convergence, making the NtoN communication strategy the most effective communication scheme under the examined experimental configuration.

Table 6. Comparison of average objective function evaluations for different propagation methods using 10 subpopulations.

FUNCTION	1to1	1toN	Nto1	NtoN
BUCHÉ RASTRIGIN_25	19722	7300	9020	4866
BUCHÉ RASTRIGIN_50	26858	10807	12167	5874
BUCHÉ RASTRIGIN_100	39765	13431	16701	6429
BUCHÉ RASTRIGIN_150	48431	14223	17452	4715
SHARP RIDGE_25	15634	7192	8121	4795
SHARP RIDGE_50	16223	7557	8315	5338
SHARP RIDGE_100	16237	8187	9257	5644
SHARP RIDGE_150	16516	7584	9072	5640
ZAKHAROV_25	8706	5610	6264	4827
ZAKHAROV_50	12481	7000	7654	5179
ZAKHAROV_100	19088	8844	10881	4803
ZAKHAROV_150	21919	8894	10539	4919

Figure 5 illustrates the effect of the four communication strategies (1to1, 1toN, Nto1 and NtoN) using a fixed configuration of 10 subpopulations. Lower numbers of objective function evaluations indicate better optimization performance. The figure shows that the NtoN communication strategy consistently achieves the lowest median number of objective function evaluations and exhibits the smallest variability among the examined strategies, whereas the 1to1 strategy generally requires the highest computational effort. A Friedman

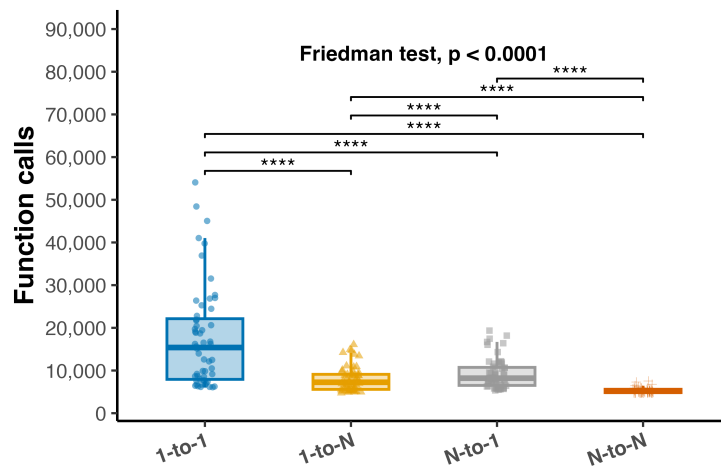


Figure 5. Impact of Communication Strategies with 10 Subpopulations.

test ($p < 0.0001$) indicates statistically significant differences among the four communication strategies. Pairwise Wilcoxon signed-rank tests further confirm these differences using the conventional significance notation (ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$). Overall, these results indicate that the NtoN communication strategy provides the most effective communication scheme under the examined experimental configuration.

3.2.4. Effect of the Adaptive Factor

To evaluate the effectiveness of the proposed adaptive factor, an ablation study was conducted by comparing the proposed square-root schedule $A = \sqrt{\text{iter}/T}$ with three alternative formulations: a constant schedule $A = 1$, an exponential schedule and a sigmoid-based schedule. The Büche Rastrigin function was selected because its multimodal landscape provides a challenging test for examining the balance between stochastic exploration and guidance toward the best solution. All four configurations were evaluated under identical experimental conditions across dimensions ranging from 25 to 500. The average number of objective function evaluations is reported in Table 7.

Table 7. Average number of objective function evaluations obtained using different adaptive-factor formulations on the Büche Rastrigin function.

FUNCTION	$A = \sqrt{\text{iter}/T}$	$A = 1$	$A_{\text{exponential}}$	A_{Sigmoid}
BUCHERASTRIGIN_25	4866	4904	4655	4652
BUCHERASTRIGIN_50	5874	6137	6133	6106
BUCHERASTRIGIN_100	6429	6991	6405	6774
BUCHERASTRIGIN_150	4715	6513	6131	6081
BUCHERASTRIGIN_200	5853	6300	5221	5576
BUCHERASTRIGIN_250	5678	5843	5460	5552
BUCHERASTRIGIN_300	4934	5114	5512	5518
BUCHERASTRIGIN_350	5480	5580	5135	5134
BUCHERASTRIGIN_400	5154	5188	5143	5138
BUCHERASTRIGIN_450	5221	5321	5061	5094
BUCHERASTRIGIN_500	5254	5362	5077	5077

Table 7 summarizes the average number of objective function evaluations obtained using four factor formulations. The proposed square-root schedule required fewer objective function evaluations than the constant formulation $A = 1$ across all examined dimensions. Averaged over the eleven benchmark instances, the square-root schedule required approxi-

mately 5,405 evaluations, compared with 5,750 for the constant formulation, corresponding to a reduction of approximately 6.0%. The largest improvement over $A = 1$ was observed for the 150-dimensional instance, where the number of evaluations decreased from 6,513 to 4,715 (approximately 27.6%). However, the exponential and sigmoid schedules achieved lower evaluation counts than the square-root formulation in several dimensions. Therefore, no single adaptive schedule was uniformly superior across all examined instances. Nevertheless, the results demonstrate that adaptive formulations generally provide advantages over the constant setting, although their relative effectiveness depends on the problem dimension.

3.3. Performance Comparison with Existing Optimization Algorithms

Table 8 presents a comparative evaluation of the proposed optimization algorithm against a set of well-established metaheuristic algorithms, including GA [29,30], DE [31,32], GWO [33,34], ParallelDE [35], SAOP [36], BGWO [37,38] and WOA [39,40] based on the average number of objective function evaluations required for convergence. Lower values indicate higher computational efficiency. Overall, the proposed method demonstrates competitive performance across the considered benchmark suite while maintaining a relatively low computational cost over different problem dimensions. Across all benchmark instances, the proposed method requires a total of 297,299 objective function evaluations, compared with 777,899, 762,504, 772,866, 314,144, 263,370, 724,617 and 371,962 for GA, DE, GWO, ParallelDE, SAOP, BGWO and WOA respectively. Although SAOP achieves a lower cumulative number of objective function evaluations, the proposed method consistently exhibits competitive performance across the benchmark suite and maintains low computational cost for the majority of benchmark instances. These results indicate that the proposed cooperative framework achieves an effective balance between convergence speed and computational efficiency under the considered experimental conditions.

Table 8. Average objective function evaluations obtained by the proposed method and competing metaheuristic algorithms.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO	WOA	PROPOSED
BUCHE RASTRIGIN_25	11233(0.93)	11417(0.93)	4214	5592	2183(0.93)	1989(0.93)	3009(0.97)	4866
BUCHE RASTRIGIN_50	17589(0.57)	21873(0.57)	13509(0.90)	5610	4657(0.57)	4033(0.57)	14163(0.70)	5874
BUCHE RASTRIGIN_100	32881(0.30)	39401(0.30)	23867(0.80)	5622	7806(0.30)	7564(0.30)	7293(0.97)	6429
BUCHE RASTRIGIN_150	36444(0.40)	48017(0.40)	22139(0.87)	5611	7731(0.40)	6472(0.40)	5699	4715
SHARP RIDGE_25	11324	11800	4098	5596	2124	4087	2110	4795
SHARP RIDGE_50	11454	12866	16581	5612	3106	18454	5907	5338
SHARP RIDGE_100	12105	12682	29123	5624	3995	28790	8936	5644
SHARP RIDGE_150	12266	12231	25399	5614	4055	24923	8022	5640
ZAKHAROV_25	5760	4939	8426	5598	2177	9075	10581	4827
ZAKHAROV_50	15253	8154	23034	5611	3083	27447	25061	5179
ZAKHAROV_100	36693	12572	3329	5623	3878	5763	9463	4803
ZAKHAROV_150	38858	14744	3251	5613	4341	3333	11943	4919

Figure 6 compares the examined optimization algorithms in terms of the number of objective function evaluations required to achieve convergence. Lower values indicate better computational efficiency. The proposed method exhibits one of the lowest median numbers of objective function evaluations while maintaining a relatively small variability across the benchmark problems. A Kruskal–Wallis test ($p < 0.0001$) indicates statistically significant differences among the compared algorithms. Overall, the results suggest that the proposed method provides competitive optimization performance and a consistent convergence behavior under the considered experimental conditions.

3.4. High-Dimensional Benchmark Problems

To further investigate the scalability of the proposed framework, additional experiments were conducted on two high-dimensional benchmark functions with dimensions ranging from 50 to 500. Tables 9 and 10 summarize the obtained results. Overall, the

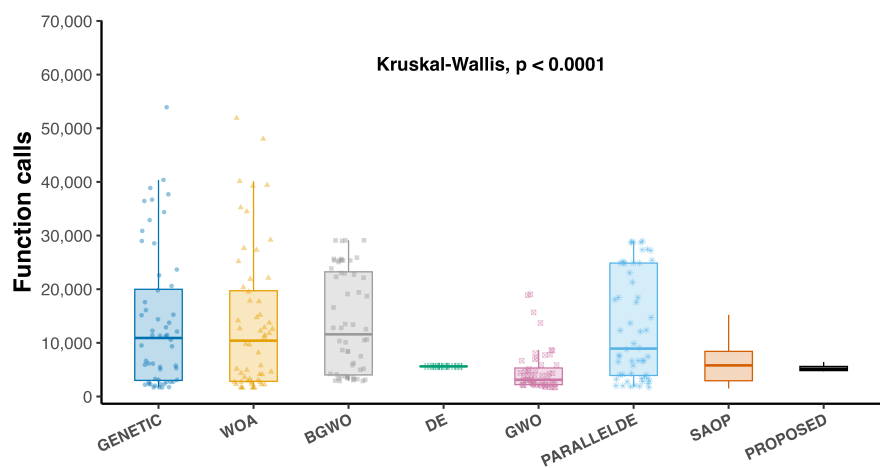


Figure 6. The proposed method in comparison with others.

relative performance of the compared algorithms depends on both the benchmark function and the problem dimensionality. The proposed method demonstrates competitive performance against both the classical metaheuristics: GA, DE, WOA, BICCA [41] SHADE-ILS [43] MLSHADE-SPA [42] and RDEx [44]. Although some competing methods achieve lower computational costs on specific benchmark instances, the proposed method remains competitive across a broad range of problem dimensions and benchmark characteristics. These results indicate that the proposed cooperative framework maintains its optimization capability as the problem dimensionality increases under the considered experimental conditions.

Table 9. Average objective function evaluations for high-dimensional optimization problems.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO
KATSUURA_50	21018	22597	2895	5595	3387	1746
KATSUURA_100	28941	28824	2923	5743	2834	4076
KATSUURA_200	31090	27304	2923	5583	2719	1738
KATSUURA_300	35067	26908	4058	5617	2719	1743
KATSUURA_400	36100	28200	2909	5611	2810	1743
KATSUURA_500	35002	28053	3200	5604	2880	1743

Table 10. Average objective function evaluations for high-dimensional optimization problems.

FUNCTION	WOA	PROPOSED	BICCA	SHADE_ILS	MLSHADESPA	RDEx
KATSUURA_50	1509	6020	10126	89853	48773	1328 (0.33)
KATSUURA_100	1523	5857	20207	92019	49012	1240 (0.66)
KATSUURA_200	1523	6064	40945	94328	45002	1250 (0.67)
KATSUURA_300	9343	6017	61107	96005	47900	2029
KATSUURA_400	1516	6021	80713	95079	48102	1142
KATSUURA_500	9323	6059	81723	98001	48307	2030

The mean objective-function values and the corresponding standard deviations reported in tables 11 and 12 provide further evidence regarding the accuracy and stability of the examined methods.

Table 11. Mean objective function values (standard deviation) for high-dimensional optimization problems.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO
KATSUURA_50	0(0)	0(0)	2.63(4.77)	0(0)	4.22(4.54)	4.45 (7.11)
KATSUURA_100	0(0)	0(0)	5.95(13.19)	0(0)	3.78(7.56)	5.25 (10.56)
KATSUURA_200	0(0)	0(0)	1.99(5.99)	0(0)	3.12(6.53)	1.99 (5.99)
KATSUURA_300	0(0)	0(0)	2.57(5.16)	0(0)	2.99(5.23)	1.27 (6.23)
KATSUURA_400	0(0)	0(0)	1.99(3.98)	0(0)	2.33(4.25)	1.21 (5.47)
KATSUURA_500	0(0)	0(0)	1.52(1.23)	0(0)	2.01(3.99)	1.09 (4.39)

Table 12. Mean objective function values (standard deviation) for high-dimensional optimization problems.

FUNCTION	WOA	PROPOSED	BICCA	SHADE_ILS	MLSHADESPA	RDEx
KATSUURA_50	0.0002 (0.0007)	3.4×10^{-108} (1.7×10^{-107})	0.04 (0.005)	0(0)	0(0)	3.39 (3.93)
KATSUURA_100	7.3×10^{-07} (1.8×10^{-06})	7.6×10^{-109} (4.1×10^{-108})	0.057 (0.007)	0(0)	0(0)	4.82 (6.81)
KATSUURA_200	1.2×10^{-08} (3.5×10^{-08})	1.7×10^{-92} (9.5×10^{-92})	0.04 (0.006)	0(0)	0(0)	8.68 (12.28)
KATSUURA_300	6.6×10^{-07} (4.6×10^{-08})	1.9×10^{-87} (8.1×10^{-87})	0.04 (0.004)	0(0)	0(0)	16.5 (0)
KATSUURA_400	6.0×10^{-06} (4.2×10^{-08})	5.2×10^{-91} (1.5×10^{-90})	0.04 (0.003)	0(0)	0(0)	0 (0)
KATSUURA_500	5.3×10^{-06} (3.6×10^{-08})	2.7×10^{-97} (6.4×10^{-96})	0.03 (0.002)	0(0)	0(0)	7.02 (0)

3.5. Parallel Performance Analysis

3.5.1. Wall-Clock Execution Time Analysis

Table 13 reports the average wall-clock execution time of the proposed PEAO framework for different numbers of subpopulations. Overall, the configurations with 10 and 20 subpopulations consistently achieved the lowest execution times across the examined benchmark problems, whereas the configurations with 5 and 30 subpopulations generally require longer execution times. These results indicate that a moderate number of subpopulations provides a more effective balance between workload distribution and coordination overhead. Although the relative differences depend on the benchmark function and problem dimensionality, the 10-subpopulation configuration consistently provides competitive execution times across the considered experiments. Overall, the results suggest that increasing the number of subpopulations beyond a moderate level does not necessarily improve runtime performance under the examined experimental conditions.

Table 13. Average wall-clock execution time (in seconds) of PEO using different numbers of clusters.

FUNCTION	5 CLUSTER	10 CLUSTER	20 CLUSTER	30 CLUSTER
ATTRACTIVESECTOR_100	5.479	5.321	5.52	6.23
ATTRACTIVESECTOR_200	7.942	7.623	7.881	8.729
ATTRACTIVESECTOR_300	8.417	7.995	8.259	9.379
ATTRACTIVESECTOR_400	10.895	9.669	9.919	11.695
ATTRACTIVESECTOR_500	14.311	13.033	13.282	15.379
ROSENBROCK_100	11.224	5.521	5.565	10.802
ROSENBROCK_200	38.821	7.824	7.81	32.234
ROSENBROCK_300	73.06	8.192	14.805	73.237
ROSENBROCK_400	191.966	10.169	10.036	182.922
ROSENBROCK_500	197.68	13.571	13.395	203.496

3.5.2. Speedup Analysis

To evaluate the parallel implementation, experiments were conducted using 1, 5 and 10 processing threads while maintaining a fixed configuration of 10 subpopulations. Table 14 reports the corresponding execution times, speedup values, and parallel efficiencies. The results show modest improvements in execution time as the number of processing threads increases, while the achieved speedup remains close to unity across the examined benchmark instances. Furthermore, parallel efficiency decreases with the number of processing threads, indicating limited scalability of the proposed implementation under the considered experimental conditions.

Table 14. Speedup and parallel efficiency of the proposed PEO for different numbers of processing threads (10 subpopulations).

FUNCTION	Threads	Time (s)	Speedup	Parallel Efficiency (%)
ROSENBROCK_100	1	9.29	1.000	100.0
ROSENBROCK_100	5	8.75	1.062	21.23
ROSENBROCK_100	10	8.7	1.068	10.68
ROSENBROCK_200	1	29.8	1.000	100.0
ROSENBROCK_200	5	28.5	1.046	20.91
ROSENBROCK_200	10	28.43	1.048	10.48
ROSENBROCK_300	1	72.143	1.000	100.0
ROSENBROCK_300	5	66.652	1.082	21.65
ROSENBROCK_300	10	65.33	1.104	11.04
ROSENBROCK_400	1	187.24	1.000	100.0
ROSENBROCK_400	5	182.445	1.026	20.53
ROSENBROCK_400	10	177.36	1.056	10.56
ROSENBROCK_500	1	270.03	1.000	100.0
ROSENBROCK_500	5	269.343	1.003	20.05
ROSENBROCK_500	10	265.08	1.019	10.19

3.6. Effect of Communication Strategies on Function Evaluations and Execution Time

To further investigate the influence of communication strategies. Figures 7, 8, 9 and 10 present the average number of objective function evaluations and execution time obtained by the three communication strategies for the Attractive Sector and Rosenbrock benchmark functions over problem dimensions ranging from 100 to 500. The results indicate that the three strategies exhibit similar scalability as the problem dimension increases. Nevertheless, Strategy 2 consistently requires the lowest number of objective function evaluations in most problem instances, while also achieving the lowest or nearly lowest execution time. Although the observed differences are relatively small, these findings suggest that Strategy 2 provides a slightly more efficient balance between information

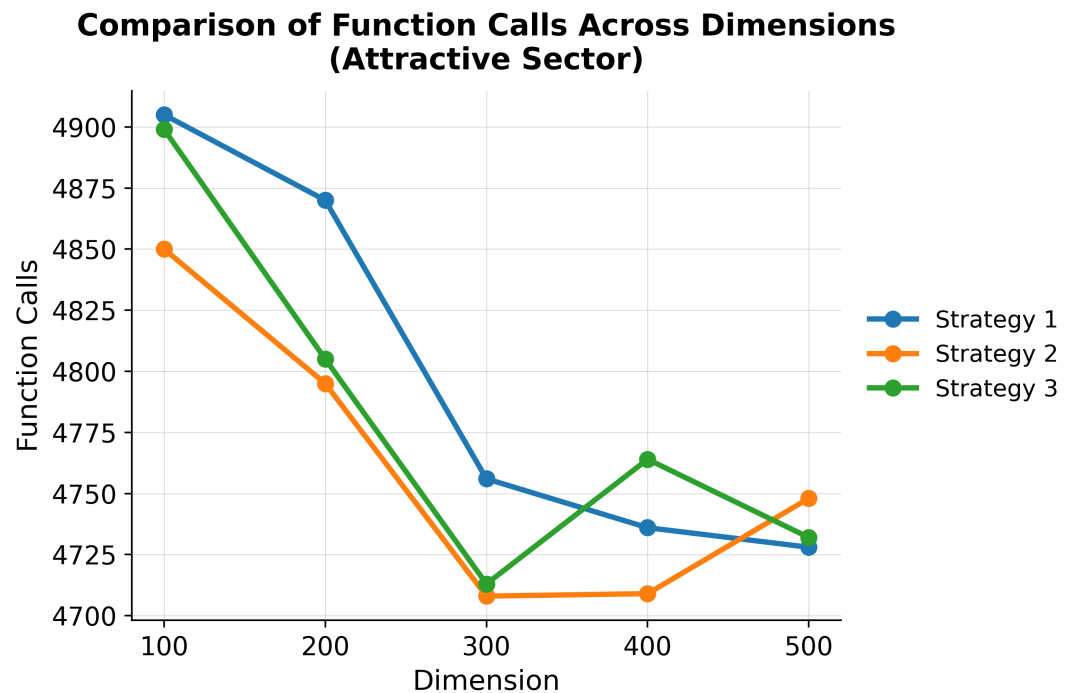


Figure 7. Comparison of objective function evaluations across problem dimensions for the three communication strategies on the Attractive Sector benchmark function.

exchange and computational overhead, making it the preferred communication strategy for the proposed PEA framework.

3.7. Real-World Optimization Problems

To further assess the practical applicability of the proposed method, three real-world optimization problems with different characteristics were considered: Dynamic Economic Dispatch 1 [45], Dynamic Economic Dispatch 2 [45] and Electricity Transmission Pricing [46,47]. Table 15 summarizes the mathematical formulation, dimensionality, variable types, constraints and key optimization characteristics of each problem. All algorithms were evaluated under identical experimental conditions using 30 independent runs, and the obtained results are reported in Tables 16 and 17.

Table 15. The real world benchmark functions used in the conducted experiments.

Problem	Formula	Dim	Constraints/Bounds
Electricity Transmission Pricing	$\min_x f(x) = \sum_{i=1}^{N_g} \left(\frac{C_i^{gen}}{p_i^{gen}} - R_i^{gen} \right)^2 + \sum_{j=1}^{N_d} \left(\frac{C_j^{load}}{p_j^{load}} - R_j^{load} \right)^2$ $\sum_j GD_{i,j} + \sum_j BT_{i,j} = p_i^{gen}, \quad \forall i$ $\sum_i GD_{i,j} + \sum_i BT_{i,j} = p_j^{load}, \quad \forall j$ $GD_{i,j}^{max} = \min(p_i^{gen} - BT_{i,j}, p_j^{load} - BT_{i,j})$	126	$GD_{i,j} \in [0, GD_{i,j}^{max}]$
Dynamic Economic Dispatch 1	$\min_{\mathbf{P}} f(\mathbf{P}) = \sum_{t=1}^{24} \sum_{i=1}^5 (a_i P_{i,t}^2 + b_i P_{i,t} + c_i)$ $p_i^{min} \leq P_{i,t} \leq p_i^{max}, \quad \forall i = 1, \dots, 5, t = 1, \dots, 24$ $\sum_{i=1}^5 P_{i,t} = D_t, \quad \forall t = 1, \dots, 24$ $P_{min} = [10, 20, 30, 40, 50]$ $P_{max} = [75, 125, 175, 250, 300]$	120	$p_i^{min} \leq P_{i,t} \leq p_i^{max}$
Dynamic Economic Dispatch 2	$\min_{\mathbf{P}} f(\mathbf{P}) = \sum_{t=1}^{24} \sum_{i=1}^9 (a_i P_{i,t}^2 + b_i P_{i,t} + c_i)$ $p_i^{min} \leq P_{i,t} \leq p_i^{max}, \quad \forall i = 1, \dots, 9, t = 1, \dots, 24$ $\sum_{i=1}^5 P_{i,t} = D_t, \quad \forall t = 1, \dots, 24$ $P_{min} = [150, 135, 73, 60, 73, 57, 20, 47, 20]$ $P_{max} = [470, 460, 340, 300, 243, 160, 130, 120, 80]$	216	$p_i^{min} \leq P_{i,t} \leq p_i^{max}$

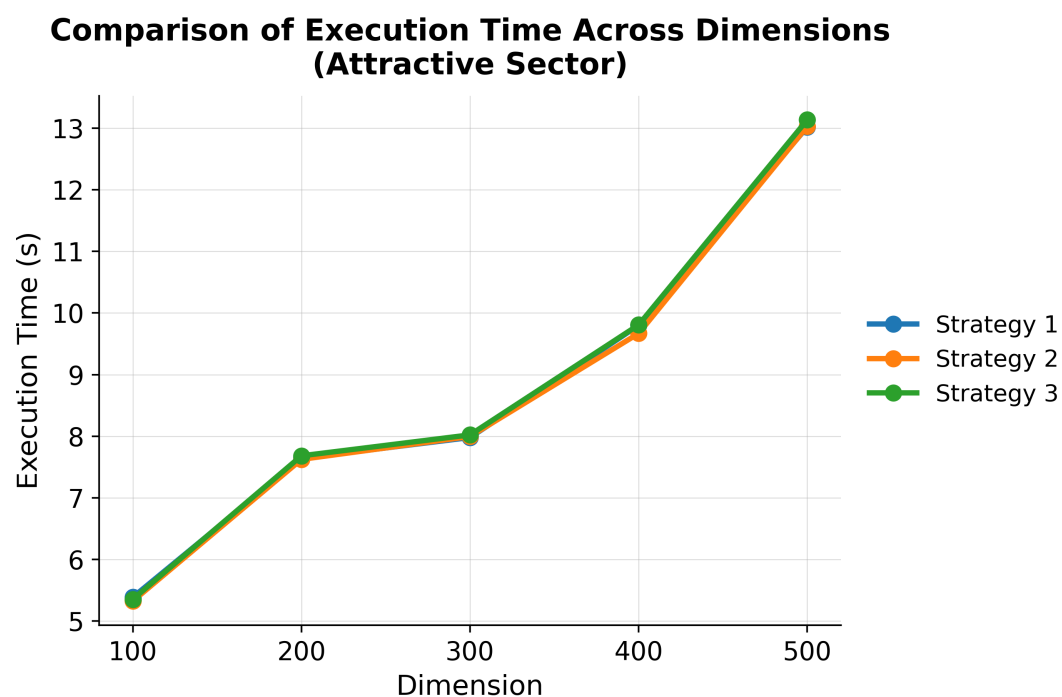


Figure 8. Comparison of execution time across problem dimensions for the three communication strategies on the Attractive Sector benchmark function.

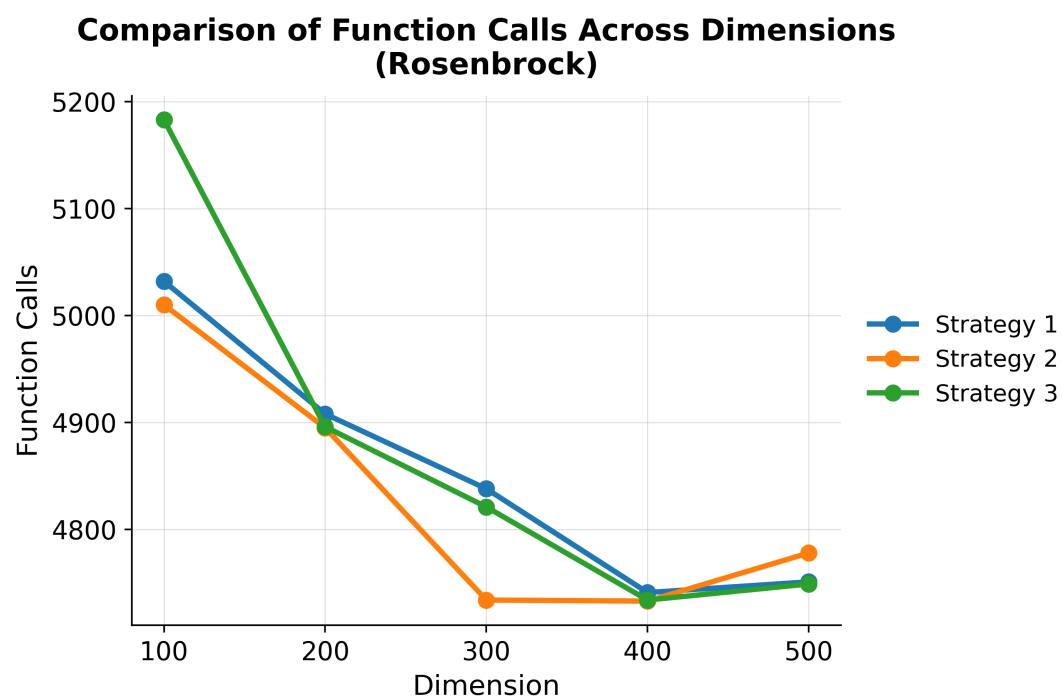


Figure 9. Comparison of objective function evaluations across problem dimensions for the three communication strategies on the Rosenbrock benchmark function.

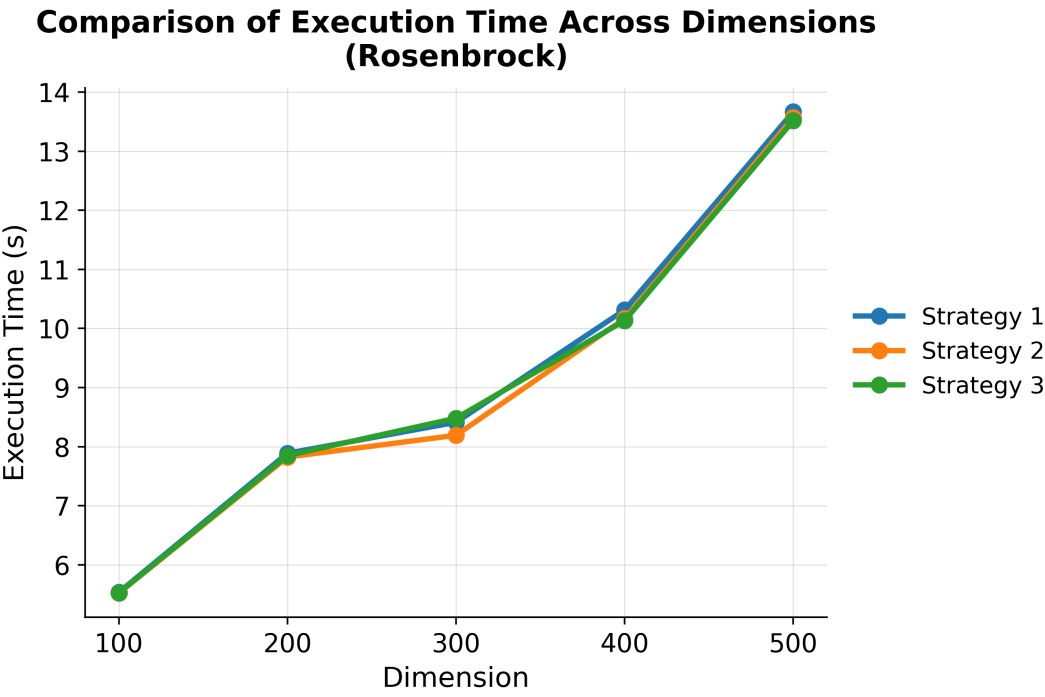


Figure 10. Comparison of execution time across problem dimensions for the three communication strategies on the Rosenbrock benchmark function.

Table 16. Optimization results on the considered real-world optimization problems mean (std) over 30 independent runs.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO
Dynamic Economic Dispatch 1	130642.67 (7.3 × 10 ^{−06})	130642 (3.4 × 10 ^{−05})	130642 (0)	130642.68 (0.002)	130642.67 (0.0003)	(130642.67) (0.0004)
Dynamic Economic Dispatch 2	164987.44 (97)	165175 (93.83)	165626 (206.63)	165845.79 (233.41)	165415.35 (243)	165622.96 (204.105)
Electricity Transmission Pricing	4.5 (7.2 × 10 ^{−10})	4.53 (8.51 × 10 ^{−10})	4.53 (3.31 × 10 ^{−10})	4.53 (9.26 × 10 ^{−07})	4.53 (4.02 × 10 ^{−10})	4.53 (7.74 × 10 ^{−07})

Table 17. Optimization results on the considered real-world optimization problems mean (std) over 30 independent runs.

FUNCTION	WOA	PROPOSED	BICCA	SHADE_ILS	MLSHADESPA	RDEx
Dynamic Economic Dispatch 1	130642.68 (0.0004)	130642.67 (3.3 × 10 ^{−05})	130642.68 (0.007)	130642.67 (1 × 10 ^{−06})	130642.67 (0.0002)	130642.69 (0.0003)
Dynamic Economic Dispatch 2	165678.26 (264.61)	165919.21 (210)	165608.63 (87.47)	165022.03 (214)	178853.01 (453)	165678.2 (52.97)
Electricity Transmission Pricing	4.53 (7.05 × 10 ^{−07})	4.53 (1.86 × 10 ^{−09})	4.53 (8.05 × 10 ^{−06})	4.53 (5.80 × 10 ^{−10})	4.53 (7.69 × 10 ^{−09})	4.53 (1.38 × 10 ^{−08})

Tables 16 and 17 report the mean and standard deviation of the best objective-function values obtained over 30 independent runs. For Dynamic Economic Dispatch 1, all compared algorithms converged to very similar objective values with negligible variability, indicating the relatively low difficulty of this instance. In contrast, Dynamic Economic Dispatch 2 exhibited greater variability among the obtained solutions, reflecting its higher optimization complexity. For the Electricity Transmission Pricing, all methods consistently achieved comparable objective values with very small standard deviations, demonstrating stable optimization performance across repeated executions. Overall, the proposed PEO produced competitive results on all three real-world optimization problems.

3.8. Statistical and Trial-Based Performance Evaluation

To evaluate the proposed PEO against the recent state-of-the-art algorithms RDE [49] and RDEx [44], comparative experiments were conducted using the trial-based dominance methodology [48]. Function F2 was excluded from the experimental evaluation due to its reported unstable behavior. Consequently, all reported results correspond to the remaining 29 benchmark functions [50]. This methodology simultaneously evaluates solution quality and convergence speed through the computation of U-scores. Table 18 presents the per-problem U-scores and the corresponding rankings for each benchmark function, whereas table 19 summarizes the overall ranking obtained by aggregating the U-scores over all benchmark functions.

Table 18. Per-problem U-scores and corresponding rankings of PEO, RDE and RDEx on the selected CEC 2017 benchmark functions.

FUNCTION	PROPOSED U-SCORE	PROPOSED RANK	RDE U-SCORE	RDE RANK	RDEx U-SCORE	RDEx RANK
cec2017f1 [d=30]	0	3	879	2	996	1
cec2017f3 [d=30]	1250	1	533	2	92	3
cec2017f4 [d=30]	1196	1	348	2	332	3
cec2017f5 [d=30]	0	3	1136	1	739	2
cec2017f6 [d=30]	0	3	748	2	1127	1
cec2017f7 [d=30]	0	3	1063	1	812	2
cec2017f8 [d=30]	0	3	1181	1	694	2
cec2017f9 [d=30]	0	3	904	2	971	1

Table 18 presents the per-problem U-scores and the corresponding rankings of PEO, RDE and RDEx on the selected CEC 2017 benchmark functions. The U-scores were computed according to the trial-based dominance methodology proposed by Price et al. [48], which simultaneously considers both convergence speed and solution accuracy. The results indicate that the recent state-of-the-art algorithms RDE and RDEx consistently achieve higher U-scores than PEO on most benchmark functions, highlighting their superior overall performance on this benchmark suite.

Table 19. Overall ranking based on the aggregated U-scores and rank sums.

Algorithm	Ranking	Sum (U-score)	RS (rank-sum)
RDE	1	28018	39
RDEx	2	20078	58
PEAO	3	6280	77

Table 19 summarizes the overall ranking obtained by aggregating the per-problem U-scores over all benchmark functions. According to this evaluation, RDE achieves the highest overall U-score and is ranked first, followed by RDEx, while PEO is ranked third. These results are consistent with the reported performance of RDE and RDEx as top-performing algorithms in the IEEE CEC 2025 competition.

4. Discussion

4.1. Effect of the Cooperative Cluster Structure

The experimental results indicate that the behavior of PEO is influenced by the number of cooperating subpopulations. Under the examined benchmark functions and parameter settings, configurations with 10 and 20 subpopulations generally required fewer objective function evaluations than configurations with 5 or 30 subpopulations. These findings suggest that a moderate population decomposition provides an effective balance between distributed exploration and local refinement. However, this observation should not be interpreted as evidence that 10 or 20 subpopulations are universally optimal, since the most appropriate configuration may depend on the optimization problem, dimensionality, population size and computing platform.

4.2. Impact of the Propagation Mechanism

The comparison between the configurations with and without propagation showed that enabling propagation consistently reduced the number of objective function evaluations. The paired Wilcoxon signed-rank test confirmed that this difference was statistically significant ($p < 0.0001$). These findings suggest that exchanging promising solutions among subpopulations improves search efficiency by reducing redundant exploration and accelerating convergence. Since objective function evaluations do not directly measure communication overhead, these results should be interpreted together with the wall-clock execution time analysis.

4.3. Analysis of Communication Strategies

The comparison of the four communication strategies indicates that the information-sharing topology has a measurable influence on optimization performance. Under the examined configuration, the N-to-N strategy generally achieved the lowest objective function evaluation counts and the smallest variability. The Friedman test indicated statistically significant differences among the evaluated communication schemes. Nevertheless, the observed advantage of the N-to-N topology is limited to the benchmark functions, population size and number of subpopulations considered in this study.

4.4. Performance on Practical Optimization Problems

The experiments on the practical optimization problems provide additional evidence regarding the behavior of the evaluated methods under realistic optimization settings. Although the problems differ substantially from the synthetic benchmark functions, the proposed method maintained competitive performance in terms of objective function evaluations and solution quality.

4.5. Comparative Performance Evaluation

The comparative study indicates that PEAO achieved competitive performance across the examined benchmark suite. It obtained relatively low objective function evaluation counts on many problems while maintaining stable performance across different dimensions. The Kruskal–Wallis test confirmed statistically significant differences among the compared algorithms, indicating that their performance distributions are not identical. However, these statistical results should be interpreted together with the function-by-function results, since no single algorithm consistently achieved the best performance on every benchmark.

4.6. Overall Assessment

Overall, the experimental results indicate that the proposed PEAO framework provides competitive optimization performance under the examined benchmark functions and experimental settings. The sensitivity analyses further suggest that population decomposition, propagation, communication strategy and adaptive search control all influence optimization efficiency. Although several competing algorithms achieved superior performance on individual problems, the proposed framework maintained a competitive and consistent performance profile across a broad range of optimization scenarios.

5. Conclusions

This study presented the Parallel Enzyme Action Optimizer (PEAO), a cooperative population-based optimization framework that extends the original EAO through parallel subpopulation evolution, controlled propagation, alternative communication topologies, adaptive search control, optional local refinement and a convergence-based stopping crite-

rion. The proposed framework was designed to maintain population diversity, facilitate information exchange among subpopulations and balance exploration and exploitation throughout the optimization process. The experimental analysis demonstrated that several design components substantially influence the behavior of the proposed framework. In particular, moderate numbers of subpopulations generally require fewer objective function evaluations than smaller or larger decompositions, while enabling propagation consistently improved evaluation efficiency. Among the examined communication topologies, the N-to-N configuration generally required the fewest objective function evaluations when 10 subpopulations were employed. The adaptive square-root schedule also provided stable and competitive performance across the examined problem dimensions. The comparative experiments indicate that PEAO achieved competitive performance across the considered benchmark problems, including high-dimensional instances with up to 500 decision variables. The proposed framework generally required comparatively few objective function evaluations while maintaining competitive final-solution quality across a wide range of benchmark functions. Nevertheless, several competing algorithms achieved superior performance on individual functions or dimensional configurations, indicating that no single optimization algorithm consistently outperforms all alternatives. The experiments on the practical optimization problems further demonstrate that the proposed framework maintains competitive performance beyond synthetic benchmark functions. However, since these problems were evaluated only in their original fixed-dimensional formulations, they should not be interpreted as evidence of scalability but rather as an assessment of practical optimization performance under the examined conditions. Overall, the experimental findings suggest that the proposed cooperative framework provides a competitive balance between search efficiency, solution quality and robustness under the examined experimental settings. The conclusions are limited to the benchmark suite, parameter settings, stopping criterion, implementation and computing platform considered in this study. Future work will investigate self-adaptive communication and propagation policies, dynamic management of subpopulations, derivative-free alternatives to the optional local refinement stage and a more comprehensive parallel scalability analysis using different numbers of processing threads. Additional validation on recent benchmark suites and a broader collection of large-scale and real-world optimization problems will also be considered to further assess the generality of the proposed framework.

Author Contributions: G.K., V.C. and I.G.T. conceived of the idea and the methodology, and G.K. and V.C. implemented the corresponding software. G.K. conducted the experiments, employing objective functions as test cases, and the comparative experiments. I.G.T. performed the necessary statistical tests. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A Appendix

This appendix provides the complete numerical results of the experimental analyses presented in the main manuscript. To improve readability, the main text reports summarized results, representative examples and the corresponding statistical analyses, whereas the following tables contain the complete results for all benchmark functions, dimensions, algorithms and experimental configurations.

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Appendix A.1 Effect of the Number of Subpopulations

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Table A1. Complete comparison of different numbers of subpopulations in terms of average objective function evaluations over 30 independent runs. Values in parentheses denote the success rate.

FUNCTION	5 CLUSTER	10 CLUSTER	20 CLUSTER	30 CLUSTER
ATTRACTIVE_SECTOR_25	4796	4701	4681	4749
ATTRACTIVE_SECTOR_50	4958	4873	4737	4803
ATTRACTIVE_SECTOR_100	5104	4964	4757	4813
ATTRACTIVE_SECTOR_150	4979	4927	4750	4805
BUCHERASTRIGIN_25	6298	4866	4807	5917
BUCHERASTRIGIN_50	8368	5874	5679	7336
BUCHERASTRIGIN_100	10352	6429	6188	9071
BUCHERASTRIGIN_150	10585	4715	5934	9771
DISCUS_25	4847	4905	4687	4801
DISCUS_50	5031	4994	4777	4871
DISCUS_100	5198	5083	4806	4891
DISCUS_150	5066	4818	4789	4844
DIFFERENTPOWERS_25	6688	5575	4706	6204
DIFFERENTPOWERS_50	8919	6784	4900	6803
DIFFERENTPOWERS_100	11752	5957	4702	7611
DIFFERENTPOWERS_150	11616	5911	4809	8720
ELLIPSOIDAL_25	5352	4773	4749	5229
ELLIPSOIDAL_50	6513	5595	5387	6339
ELLIPSOIDAL_100	9340	7303	6708	9017
ELLIPSOIDAL_150	9934	7541	7927	8680
GALLAGHER21_25	5629	4773	4752	4876
GALLAGHER21_50	5652	5277	5105	5133
GALLAGHER21_100	5587	5305	5287	5515(0.93)
GALLAGHER21_150	5633	5298	5225(0.97)	5175
GALLAGHER101_25	4960	4767	4741	4865
GALLAGHER101_50	5617	5238	5117	5180
GALLAGHER101_100	6165	5452	5322	5527
GALLAGHER101_150	5650	5217	5063	5439
GRIEWANK_25	6115	4833	5157	5730
GRIEWANK_50	6018	5557	5332	5689
GRIEWANK_100	6470	6246	5924	6260
GRIEWANK_150	6277	5820	5741	6121
GRIEWANK_ROSENBROCK_25	6691	4754	4821	6412
GRIEWANK_ROSENBROCK_50	7877	5299	5147	7378
GRIEWANK_ROSENBROCK_100	8150	5536	5659	7814
GRIEWANK_ROSENBROCK_150	7614	5287	5500	7883
ROSENBROCK_25	6822	4728	4686	6202
ROSENBROCK_50	8421	5083	4779	7659
ROSENBROCK_100	12122	5145	4814	8734
ROSENBROCK_150	11149	4958	4773	9992
RARSTIGIN_25	5998	4869	4756	5631
RARSTIGIN_50	7074	5675	5352	6433
RARSTIGIN_100	7851	5999	5862	7425
RARSTIGIN_150	7507	5686	5641	7129
STEP ELLIPSOIDAL_25	4670	4669	4669	4670
STEP ELLIPSOIDAL_50	4699	4699	4697	4698
STEP ELLIPSOIDAL_100	4701	4701	4701	4702
STEP ELLIPSOIDAL_150	4696	4695	4697	4696
SHARP RIDGE_25	6158	4795	4713	6099
SHARP RIDGE_50	6707	5338	4835	5898
SHARP RIDGE_100	7153	5644	4793	6017
SHARP RIDGE_150	6699	5640	4850	5930
ZAKHAROV_25	5312	4827	4721	5092
ZAKHAROV_50	6350	5179	4770	5756
ZAKHAROV_100	8384	4803	4718	6847
ZAKHAROV_150	7434	4919	4896	7827
	385708	297299	286096(0.99)	351709(0.99)

A1 reports the complete results obtained using 5, 10, 20 and 30 subpopulations for all benchmark functions and dimensions.

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Appendix A.2 Effect of the Propagation Mechanism

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Table A2. Comparison of PEA0 with propagation disabled/enabled: average objective function evaluations over 30 runs (success rate in parentheses).

FUNCTION	PROP=0	PROP=1
ATTRACTIVE_SECTOR_25	6582	4701
ATTRACTIVE_SECTOR_50	6817	4873
ATTRACTIVE_SECTOR_100	7088	4964
ATTRACTIVE_SECTOR_150	6970	4927
BUCHE_RASTRIGIN_25	58030	4866
BUCHE_RASTRIGIN_50	86188	5874
BUCHE_RASTRIGIN_100	116211	6429
BUCHE_RASTRIGIN_150	135523	4715
DISCUS_25	7111	4905
DISCUS_50	7268	4994
DISCUS_100	7668	5083
DISCUS_150	7428	4818
DIFFERENTPOWERS_25	24052	5575
DIFFERENTPOWERS_50	32492	6784
DIFFERENTPOWERS_100	47065	5957
DIFFERENTPOWERS_150	52426	5911
ELLIPSOIDAL_25	11246	4773
ELLIPSOIDAL_50	18284	5595
ELLIPSOIDAL_100	30864	7303
ELLIPSOIDAL_150	39305	7541
GALLAGHER21_25	46411	4773
GALLAGHER21_50	13166	5277
GALLAGHER21_100	24696	5305
GALLAGHER21_150	12226(0.83)	5298
GALLAGHER101_25	69639	4767
GALLAGHER101_50	81366	5238
GALLAGHER101_100	70335	5452
GALLAGHER101_150	55891	5217
GRIEWANK_25	16507	4833
GRIEWANK_50	10253	5557
GRIEWANK_100	10034	6246
GRIEWANK_150	9414	5820
GRIEWANK_ROSENBROCK_25	24232	4754
GRIEWANK_ROSENBROCK_50	28600	5299
GRIEWANK_ROSENBROCK_100	33069	5536
GRIEWANK_ROSENBROCK_150	34478	5287
ROSENBROCK_25	22919	4728
ROSENBROCK_50	34647	5083
ROSENBROCK_100	54081	5145
ROSENBROCK_150	69505	4958
RARSTIGIN_25	51593	4869
RARSTIGIN_50	62684	5675
RARSTIGIN_100	69952	5999
RARSTIGIN_150	66543	5686
STEP_ELLIPSOIDAL_25	16205	4669
STEP_ELLIPSOIDAL_50	14935	4699
STEP_ELLIPSOIDAL_100	14030	4701
STEP_ELLIPSOIDAL_150	13747	4695
SHARP RIDGE_25	18646	4795
SHARP RIDGE_50	19932	5338
SHARP RIDGE_100	20959	5644
SHARP RIDGE_150	21109	5640
ZAKHAROV_25	10499	4827
ZAKHAROV_50	15031	5179
ZAKHAROV_100	23334	4803
ZAKHAROV_150	28647	4919
	1897933(0.99)	297299

Table A2 presents the complete comparison of PEO with and without the propagation mechanism.

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Appendix A.3 Effect of Communication Strategies

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Table A3. Complete comparison of the examined information-exchange topologies using 10 subpopulations.

FUNCTION	1to1	1toN	Nto1	NtoN
ATTRACTIVE_SECTOR_25	6083	4978	5339	4701
ATTRACTIVE_SECTOR_50	6166	5268	5498	4873
ATTRACTIVE_SECTOR_100	6157	5334	5741	4964
ATTRACTIVE_SECTOR_150	6320	5385	5797	4927
BUCHERASTRIGIN_25	19722	7300	9020	4866
BUCHERASTRIGIN_50	26858	10807	12167	5874
BUCHERASTRIGIN_100	39765	13431	16701	6429
BUCHERASTRIGIN_150	48431	14223	17452	4715
DISCUS_25	6438	5059	5582	4905
DISCUS_50	6636	5340	5713	4994
DISCUS_100	6713	5493	5688	5083
DISCUS_150	6695	5393	5799	4818
DIFFERENTPOWERS_25	20623	9105	9000	5575
DIFFERENTPOWERS_50	27698	10923	12101	6784
DIFFERENTPOWERS_100	36931	14693	16394	5957
DIFFERENTPOWERS_150	45047	16075	18178	5911
ELLIPSOIDAL_25	9841	6046	6538	4773
ELLIPSOIDAL_50	15148	7397	8320	5595
ELLIPSOIDAL_100	24463	10116	11613	7303
ELLIPSOIDAL_150	31545	11082	14318	7541
GALLAGHER21_25	9893	5747	6903	4773
GALLAGHER21_50	6252(0.80)	4736(0.87)	7365	5277
GALLAGHER21_100	8017	6159	7492	5305
GALLAGHER21_150	7008	5625	7159	5298
GALLAGHER101_25	9163	5480	6423	4767
GALLAGHER101_50	10494	6434	7581	5238
GALLAGHER101_100	12597	7330	7946	5452
GALLAGHER101_150	12135	6767	7883	5217
GRIEWANK_25	13986	7166	8057	4833
GRIEWANK_50	9204	6778	7250	5557
GRIEWANK_100	8654	6913	7424	6246
GRIEWANK_150	8560	6645	7228	5820
GRIEWANK_ROSENBROCK_25	19444	8866	9388	4754
GRIEWANK_ROSENBROCK_50	22805	8849	10687	5299
GRIEWANK_ROSENBROCK_100	25277	9533	11610	5536
GRIEWANK_ROSENBROCK_150	27005	9827	11982	5287
ROSENBROCK_25	18841	7986	8904	4728
ROSENBROCK_50	26380	9761	11855	5083
ROSENBROCK_100	41032	13803	16025	5145
ROSENBROCK_150	54071	14920	19354	4958
RARSTIGIN_25	16776	6537	8290	4869
RARSTIGIN_50	18693	8309	9526	5675
RARSTIGIN_100	20437	8799	10556	5999
RARSTIGIN_150	21729	9109	10421	5686
STEP ELLIPSOIDAL_25	7659	4923	5879	4669
STEP ELLIPSOIDAL_50	7410	4966	6052	4699
STEP ELLIPSOIDAL_100	7549	5040	6319	4701
STEP ELLIPSOIDAL_150	7657	5176	5969	4695
SHARP RIDGE_25	15634	7192	8121	4795
SHARP RIDGE_50	16223	7557	8315	5338
SHARP RIDGE_100	16237	8187	9257	5644
SHARP RIDGE_150	16516	7584	9072	5640
ZAKHAROV_25	8706	5610	6264	4827
ZAKHAROV_50	12481	7000	7654	5179
ZAKHAROV_100	19088	8844	10881	4803
ZAKHAROV_150	21919	8894	10539	4919
	982812(0.99)	446500(0.99)	518590	297299

Table A3 reports the complete results obtained using the 1-to-1, 1-to-N, N-to-1 and N-to-N information-exchange topologies with 10 subpopulations.

Appendix A.4 Complete Comparison with Competing Algorithms

Table A4. Complete comparison of PEO and the competing metaheuristic algorithms in terms of average objective function evaluations over 30 independent runs. Values in parentheses denote the success rate.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO	WOA	PROPOSED
ATTRACTIVE_SECTOR_25	2281	2234	3772	5594	1715	3239	1944	4701
ATTRACTIVE_SECTOR_50	2297	2298	12556	5611	2106	17575	4646	4873
ATTRACTIVE_SECTOR_100	2344	2273	29011	5623	2691	28735	6739	4964
ATTRACTIVE_SECTOR_150	2345	2255	25348	5614	2618	24872	6070	4927
BUCHERASTRIGIN_25	11233(0.93)	11417(0.93)	4214	5592	2183(0.93)	1989(0.93)	3009(0.97)	4866
BUCHERASTRIGIN_50	17589(0.57)	21873(0.57)	13509(0.90)	5610	4657(0.57)	4033(0.57)	14163(0.70)	5874
BUCHERASTRIGIN_100	32881(0.30)	39401(0.30)	23867(0.80)	5622	7806(0.30)	7564(0.30)	7293(0.97)	6429
BUCHERASTRIGIN_150	36444(0.40)	48017(0.40)	22139(0.87)	5611	7731(0.40)	6472(0.40)	5699	4715
DISCUS_25	2715	2612	4979	5596	1747	4266	2505	4905
DISCUS_50	2714	2701	19111	5614	2472	18438	8241	4994
DISCUS_100	2752	2693	29094	5624	2722	28740	12795	5083
DISCUS_150	2750	2671	25351	5614	2483	24876	11185	4818
DIFFERENTPOWERS_25	13758	14154	3580	5595	2456	2728	1864	5575
DIFFERENTPOWERS_50	19777	22093	10497	5612	8075	12334	4211	6784
DIFFERENTPOWERS_100	28988	29193	25884	5624	13699	27204	6439	5957
DIFFERENTPOWERS_150	34372	34490	25642	5613	15676	25413	5930	5911
ELLIPSOIDAL_25	5930	5778	3598	5593	1847	2778	1843	4773
ELLIPSOIDAL_50	10583	11023	10624	5611	2979	13691	4127	5595
ELLIPSOIDAL_100	20604	20345	25068	5623	5137	28976	6014	7303
ELLIPSOIDAL_150	28556	27643	25642	5612	5893	25171	5653	7541
GALLAGHER21_25	3118(0.93)	2867(0.93)	3214(0.93)	5595	2206(0.93)	2115(0.93)	3978(0.93)	4773
GALLAGHER21_50	3272(0.57)	3566(0.57)	6234(0.57)	5612	8557(0.57)	6590(0.57)	14831(0.57)	5277
GALLAGHER21_100	1693(0.30)	1621(0.30)	2901(0.30)	5622	1691(0.30)	8781(0.30)	1502(0.30)	5305
GALLAGHER21_150	1682(0.40)	1617(0.40)	2898(0.40)	5613	1671(0.40)	1703(0.40)	1500(0.40)	5298
GALLAGHER101_25	3083(0.93)	3030(0.93)	3106(0.93)	5593	2068	1913(0.93)	3531(0.93)	4767
GALLAGHER101_50	6150(0.57)	4617(0.57)	5873(0.57)	5602	6932	7429(0.57)	13266(0.57)	5238
GALLAGHER101_100	6656(0.30)	4340(0.30)	8356(0.30)	5606	18872	12116(0.30)	18537(0.30)	5452
GALLAGHER101_150	6122(0.40)	3236(0.40)	7496(0.40)	5601	19066	14717(0.40)	15220(0.40)	5217
GRIEWANK_25	9514	9655	3695(0.97)	5594	2237	2878(0.97)	1893	4833
GRIEWANK_50	5511	5147	10342(0.97)	5610	3132	12370(0.83)	4184	5557
GRIEWANK_100	5078	4141	22295	5623	4053	27343(0.87)	6532(0.97)	6246
GRIEWANK_150	5284	4131	25374	5612	3934	24895(0.93)	6098(0.97)	5820
GRIEWANK_ROSENBROCK_25	16098	17837	3429	5595	2259	2435	1740	4754
GRIEWANK_ROSENBROCK_50	22598	27316	8659	5612	4413	9859	3361	5299
GRIEWANK_ROSENBROCK_100	30869	35225	19410	5624	5856	21301	4784	5536
GRIEWANK_ROSENBROCK_150	37686	40126	22950	5613	6664	23047	4450	5287
ROSENBROCK_25	15148	15199	3760	5595	2217	3320	1925	4728
ROSENBROCK_50	23663	25189	12789	5612	4238	18057	4611	5083
ROSENBROCK_100	40370	39254	29103	5623	7197	28756	6786	5145
ROSENBROCK_150	53927	51902	25367	5613	8677	24893	6076	4958
RASTIGIN_25	9348(0.93)	9799(0.93)	2959(0.93)	5594	2067(0.93)	2007(0.93)	3007(0.97)	4869
RASTIGIN_50	11414(0.57)	13738(0.57)	13386(0.87)	5610	3871(0.57)	4022(0.57)	15194(0.67)	5675
RASTIGIN_100	12898(0.30)	17722(0.30)	22715(0.77)	5622	4840(0.30)	7611(0.30)	10638(0.87)	5999
RASTIGIN_150	14392(0.40)	19506(0.40)	18708(0.73)	5612	4693(0.40)	6693(0.40)	6827(0.97)	5686
STEP ELLIPSOIDAL_25	1718(0.93)	1664(0.93)	3078	5593	1806(0.93)	1831	1590	4669
STEP ELLIPSOIDAL_50	2201(0.57)	1661(0.57)	5163	5610	2369(0.57)	3615(0.90)	2273(0.93)	4699
STEP ELLIPSOIDAL_100	2028(0.30)	1629(0.30)	10101(0.97)	5622	2355(0.30)	6624(0.57)	2708(0.97)	4701
STEP ELLIPSOIDAL_150	1752(0.40)	1617(0.40)	12778(0.97)	5612	1977(0.40)	6730(0.44)	2527(0.97)	4695
SHARP RIDGE_25	11324	11800	4098	5596	2124	4087	2110	4795
SHARP RIDGE_50	11454	12866	16581	5612	3106	18454	5907	5338
SHARP RIDGE_100	12105	12682	29123	5624	3995	28790	8936	5644
SHARP RIDGE_150	12266	12231	25399	5614	4055	24923	8022	5640
ZAKHAROV_25	5760	4939	8426	5598	2177	9075	10581	4827
ZAKHAROV_50	15253	8154	23034	5611	3083	27447	25061	5179
ZAKHAROV_100	36693	12572	3329	5623	3878	5763	9463	4803
ZAKHAROV_150	38858	14744	3251	5613	4341	3333	11943	4919
	777899(0.84)	762504(0.84)	772866(0.91)	314144	263370(0.84)	724617(0.85)	371962(0.92)	297299

Table A4 provides the complete function-by-function comparison between PEO and the competing metaheuristic algorithms.

Appendix A.5 Complete experimental results using different high-dimensional optimization methods

Table A5. Average objective function evaluations on additional high-dimensional benchmark problems.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO
WEIERSTRASS_50	9569	11061	2929	5615	4125	1758
WEIERSTRASS_100	40582	13846	40419	5748	6680	40437
WEIERSTRASS_200	40794	14020	40442	5588	6900	1758
WEIERSTRASS_300	41900	14900	40573	5637	7103	40779
WEIERSTRASS_400	40500	15981	2929	5631	6800	1730
WEIERSTRASS_500	40852	16753	40559	5624	7256	1746
KATSUURA_50	21018	22597	2895	5595	3387	1746
KATSUURA_100	28941	28824	2923	5743	2834	4076
KATSUURA_200	31090	27304	2923	5583	2719	1738
KATSUURA_300	35067	26908	4058	5617	2719	1743
KATSUURA_400	36100	28200	2909	5611	2810	1743
KATSUURA_500	35002	28053	3200	5604	2880	1743

Table A6. Average objective function evaluations on additional high-dimensional benchmark problems.

FUNCTION	WOA	PROPOSED	BICCA	SHADE_ILS	MLSHADESPA	RDEx
WEIERSTRASS_50	1536	7110	10205	80083	28754	1154 (0.33)
WEIERSTRASS_100	12723	7820	20520	113280	51742	1126 (0.33)
WEIERSTRASS_200	13723	6018	40639	90127	80790	1178 (0.33)
WEIERSTRASS_300	12923	7228	60793	91340	70132	1104
WEIERSTRASS_400	1536	7185	81330	91980	90127	1010
WEIERSTRASS_500	13923	5936	90020	90140	92040	1246
KATSUURA_50	1509	6020	10126	89853	48773	1328 (0.33)
KATSUURA_100	1523	5857	20207	92019	49012	1240 (0.66)
KATSUURA_200	1523	6064	40945	94328	45002	1250 (0.67)
KATSUURA_300	9343	6017	61107	96005	47900	2029
KATSUURA_400	1516	6021	80713	95079	48102	1142
KATSUURA_500	9323	6059	81723	98001	48307	2030

Tables A5 and A6 provide the complete function-by-function comparison between PEO and the competing metaheuristic algorithms on high-dimensional problems.

Appendix A.6 Complete experimental results using mean and standard deviation for for optimization methods.

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Table A7. Mean objective function values (standard deviation) for additional high-dimensional optimization problems.

FUNCTION	GA	DE	GWO	PARALLELDE	SAOP	BGWO
WEIERSTRASS_50	0(0)	0(0)	0.54(1.43)	0(0)	0(0)	0.01 (0.02)
WEIERSTRASS_100	0(0)	0(0)	0.86(2.19)	0(0)	0(0)	0.44 (0.31)
WEIERSTRASS_200	0(0)	0(0)	3.37(3.83)	0(0)	0(0)	2.26 (2.22)
WEIERSTRASS_300	0(0)	0(0)	3.73(7.4)	0(0)	0(0)	3.93 (8.55)
WEIERSTRASS_400	0(0)	0(0)	11.85(23.73)	0(0)	0(0)	4.23 (9.11)
WEIERSTRASS_500	0(0)	0(0)	18.23(36.58)	0(0)	0(0)	4.78 (9.22)
KATSUURA_50	0(0)	0(0)	2.63(4.77)	0(0)	4.22(4.54)	4.45 (7.11)
KATSUURA_100	0(0)	0(0)	5.95(13.19)	0(0)	3.78(7.56)	5.25 (10.56)
KATSUURA_200	0(0)	0(0)	1.99(5.99)	0(0)	3.12(6.53)	1.99 (5.99)
KATSUURA_300	0(0)	0(0)	2.57(5.16)	0(0)	2.99(5.23)	1.27 (6.23)
KATSUURA_400	0(0)	0(0)	1.99(3.98)	0(0)	2.33(4.25)	1.21 (5.47)
KATSUURA_500	0(0)	0(0)	1.52(1.23)	0(0)	2.01(3.99)	1.09 (4.39)

Table A8. Mean objective function values (standard deviation) for additional high-dimensional optimization problems.

FUNCTION	WOA	PROPOSED	BICCA	SHADE_ILS	MLSHADESPA	RDEx
WEIERSTRASS_50	4.6×10^{-09} (7.5×10^{-09})	0(0)	38.67 (3.09)	24.79 (28.37)	0(0)	38.04 (276.95)
WEIERSTRASS_100	2.1×10^{-08} (2.2×10^{-08})	0(0)	105.61 (3.51)	0(0)	0(0)	105.47 (1.61)
WEIERSTRASS_200	1.6×10^{-08} (1.97×10^{-08})	0(0)	165.91 (5.98)	0(0)	0(0)	121.77 (86.12)
WEIERSTRASS_300	4.5×10^{-09} (9.1×10^{-09})	0(0)	203.89 (6.89)	0(0)	0(0)	0(0)
WEIERSTRASS_400	4.9×10^{-09} (1×10^{-08})	0(0)	253.04 (7.02)	0(0)	0(0)	0 (0)
WEIERSTRASS_500	4.3×10^{-09} (1.7×10^{-08})	0(0)	240 (6.48)	0(0)	0(0)	0 (0)
KATSUURA_50	0.0002 (0.0007)	3.4×10^{-108} (1.7×10^{-107})	0.04 (0.005)	0(0)	0(0)	3.39 (3.93)
KATSUURA_100	7.3×10^{-07} (1.8×10^{-06})	7.6×10^{-109} (4.1×10^{-108})	0.057 (0.007)	0(0)	0(0)	4.82 (6.81)
KATSUURA_200	1.2×10^{-08} (3.5×10^{-08})	1.7×10^{-92} (9.5×10^{-92})	0.04 (0.006)	0(0)	0(0)	8.68 (12.28)
KATSUURA_300	6.6×10^{-07} (4.6×10^{-08})	1.9×10^{-87} (8.1×10^{-87})	0.04 (0.004)	0(0)	0(0)	16.5 (0)
KATSUURA_400	6.0×10^{-06} (4.2×10^{-08})	5.2×10^{-91} (1.5×10^{-90})	0.04 (0.003)	0(0)	0(0)	0 (0)
KATSUURA_500	5.3×10^{-06} (3.6×10^{-08})	2.7×10^{-97} (6.4×10^{-96})	0.03 (0.002)	0(0)	0(0)	7.02 (0)

Tables A7 and A8 provide the complete function-by-function comparison between PEA0 and the competing metaheuristic algorithms on high-dimensional problems.

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Appendix A.7 Statistical and Trial-Based Performance Evaluation

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Table A9. Performance comparison of PEA0, RDE and RDE_x on selected CEC 2017 benchmark functions based on best objective value over 30 independent runs.

FUNCTION	PROPOSED	RDE	RDE _x
cec2017f1 [d=30]	100.53	100	100
cec2017f3 [d=30]	300	300	300
cec2017f4 [d=30]	400	400	400
cec2017f5 [d=30]	712.91	506.07	509.36
cec2017f6 [d=30]	644.58	600	600
cec2017f7 [d=30]	1128.04	738.1	742.65
cec2017f8 [d=30]	973.12	806.96	809.97
cec2017f9 [d=30]	1811.38	900	900
cec2017f10 [d=30]	3372.76	2682.29	3229.98
cec2017f11 [d=30]	1156.71	1101.56	1103.6
cec2017f12 [d=30]	5632.46	1215.45	1354.27
cec2017f13 [d=30]	1471.75	1302.04	1304.82
cec2017f14 [d=30]	1474.72	1421.64	1424.29
cec2017f15 [d=30]	1527.45	1502.62	1505.33
cec2017f16 [d=30]	1963.32	1614.34	1634.26
cec2017f17 [d=30]	1827.31	1730.71	1734.95
cec2017f18 [d=30]	1860.75	1800.91	1820.93
cec2017f19 [d=30]	1986.29	1903.08	1911.9
cec2017f20 [d=30]	2229.34	2027.72	2042.64
cec2017f21 [d=30]	2503.04	2308.1	2310.16
cec2017f22 [d=30]	2300	2300	2300
cec2017f23 [d=30]	2930.99	2639.01	2640.76
cec2017f24 [d=30]	2990.85	2819.6	2822
cec2017f25 [d=30]	2875.86	2886.69	2886.71
cec2017f26 [d=30]	2800	2900	3426.57
cec2017f27 [d=30]	3146.24	3172.45	3188.99
cec2017f28 [d=30]	3100	3100	3100
cec2017f29 [d=30]	3336.74	3342.88	3366.07
cec2017f30 [d=30]	3714.37	4971.7	4960.45

Table A10. U-score and rank comparison of PEA0, RDE and RDE_x on the selected CEC 2017 benchmark functions based on the trial-based dominance methodology.

FUNCTION	PROPOSED U-SCORE	PROPOSED RANK	RDE U-SCORE	RDE RANK	RDE _x U-SCORE	RDE _x RANK
cec2017f1 [d=30]	0	3	879	2	996	1
cec2017f3 [d=30]	1250	1	533	2	92	3
cec2017f4 [d=30]	1196	1	348	2	332	3
cec2017f5 [d=30]	0	3	1136	1	739	2
cec2017f6 [d=30]	0	3	748	2	1127	1
cec2017f7 [d=30]	0	3	1063	1	812	2
cec2017f8 [d=30]	0	3	1181	1	694	2
cec2017f9 [d=30]	0	3	904	2	971	1
cec2017f10 [d=30]	135	3	984	1	756	2
cec2017f11 [d=30]	25	3	1068	1	782	2
cec2017f12 [d=30]	0	3	1184	1	691	2
cec2017f13 [d=30]	0	3	1205	1	670	2
cec2017f14 [d=30]	0	3	1212	1	663	2
cec2017f15 [d=30]	0	3	1214	1	661	2
cec2017f16 [d=30]	10	3	1166	1	699	2
cec2017f17 [d=30]	0	3	1116	1	759	2
cec2017f18 [d=30]	0	3	1132	1	743	2
cec2017f19 [d=30]	0	3	1250	1	625	2
cec2017f20 [d=30]	0	3	1128	1	747	2
cec2017f21 [d=30]	0	3	1024	1	851	2
cec2017f22 [d=30]	150	3	550	2	1175	1
cec2017f23 [d=30]	0	3	1147	1	728	2
cec2017f24 [d=30]	0	3	1126	1	749	2
cec2017f25 [d=30]	1250	1	558	2	68	3
cec2017f26 [d=30]	150	3	862	2	864	1
cec2017f27 [d=30]	1250	1	593	2	32	3
cec2017f28 [d=30]	765	1	580	2	531	3
cec2017f29 [d=30]	50	3	1126	1	699	2
cec2017f30 [d=30]	50	3	1002	1	823	2
Sum / RS	6280	77	28018	39	20078	58

Tables A9 and A10 provide the complete function-by-function comparison between PEAO, RDE and RDEx.

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